

Comparative Interactome Analysis of α -arrestin Families in Human and *Drosophila*

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted

July 26, 2023 (this version)

Posted to bioRxiv

June 11, 2023

Sent for peer review

April 27, 2023

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Abstract

The α -arrestins form a large family of evolutionally conserved modulators that control diverse signaling pathways, including both G-protein-coupled receptor (GPCR-) mediated and non-GPCR mediated pathways, across eukaryotes. However, unlike β -arrestins, only a few α -arrestin targets and functions have been characterized. Here, using affinity purification and mass spectrometry, we constructed interactomes for six human and twelve *Drosophila* α -arrestins. The resulting high-confidence interactomes comprised 307 and 467 prey proteins in human and *Drosophila*, respectively. A comparative analysis of these interactomes predicted not only conserved binding partners, such as motor proteins, proteases, ubiquitin ligases, RNA splicing factors, and GTPase-activating proteins, but also those specific to mammals, such as histone modifiers and the subunits of V-type ATPase. Given the manifestation of the interaction between the human α -arrestin, TXNIP, and the histone-modifying enzymes, including HDAC2, we undertook a global analysis of transcription signals and chromatin structures that were affected by TXNIP knockdown. We found that TXNIP activated targets by blocking HDAC2 recruitment to targets, a result that was validated by chromatin immunoprecipitation assays. Additionally, the interactome for an uncharacterized human α -arrestin ARRDC5 uncovered multiple components in the V-type ATPase, which plays a key role in bone resorption by osteoclasts. Our study presents conserved and species-specific protein-protein interaction maps for α -arrestins, which provide a valuable resource for interrogating their cellular functions for both basic and clinical research.

eLife assessment

This **valuable** study provides a comparative interactome analysis of α -arrestin in human and *Drosophila*. Using **convincing** methodology that includes affinity purification/mass spectrometry, bioinformatic tools and experimental data in human cells, the authors identify biological roles of protein-protein interactions (PPI) involving α -arrestin in the two species. This study will serve as a broadly relevant resource for understanding the PPI network of human and *Drosophila* α -arrestins.

Introduction

The discovery of first arrestin protein in retinal rods contributed to a deeper understanding of photoreceptor signaling mediated by rhodopsin, which is one of the G protein-coupled receptor (GPCR) class. As its ability to arrest the GPCR signaling pathway, the protein was first named as “arrestin” (Kuhn, Hall, & Wilden, 1984 [DOI](#); Wilden, Wust, Weyand, & Kuhn, 1986 [DOI](#); Zuckerman & Cheasty, 1986 [DOI](#)). Shortly after this discovery of the first arrestin protein in the retina, another arrestin protein that specifically turns off β -adrenergic signaling was identified and named “ β -arrestin”. This arrestin-mediated termination of signaling from GPCRs is called “receptor desensitization” (Benovic, DeBlasi, Stone, Caron, & Lefkowitz, 1989 [DOI](#); Lohse, 1992 [DOI](#); Shenoy & Lefkowitz, 2011 [DOI](#)), one of crucial cellular process in maintaining cellular homeostasis and preventing overstimulation of signaling pathways. Further studies have revealed that β -arrestins regulate the receptor desensitization of other signaling pathways through ubiquitination and regulation of trafficking of various cargo molecules (Y. M. Kim & Benovic, 2002 [DOI](#); Malik & Marchese, 2010 [DOI](#); Puca & Brou, 2014 [DOI](#)).

Another class of arrestin, α -arrestin, was first studied in fungi and yeast (Andoh, Hirata, & Kikuchi, 2002 [DOI](#)) and subsequently recognized as new class of arrestins (Boase & Kelly, 2004 [DOI](#); Herranz et al., 2005 [DOI](#)). They contain characteristic arrestin domains, arrestin_N and arrestin_C, and PPXY motifs, which are unique to the α -arrestin clan. A phylogenetic study of arrestin proteins showed that α -arrestins are the ancestral class of the arrestin family and conserved from yeast to human (Alvarez, 2008 [DOI](#)). To date, six α -arrestins, arrestin domain containing protein 1 (ARRDC1), ARRDC2, ARRDC3, ARRDC4, ARRDC5, and thioredoxin-interacting protein (TXNIP), have been found to be in the human genome. These human α -arrestins were first studied in conjunction with β -arrestins in the regulation of the β 2-adrenergic receptor (β 2AR) in human cells. ARRDC3 and ARRDC4 works as an adaptor protein for the ubiquitination of β 2AR by recruiting the NEDD4 protein, an E3 ubiquitin ligase, through its conserved PPXY motifs (S. O. Han, Kommaddi, & Shenoy, 2013 [DOI](#); Nabhan, Pan, & Lu, 2010 [DOI](#); Shea, Rowell, Li, Chang, & Alvarez, 2012 [DOI](#)). In addition to their β 2AR-associated roles, α -arrestins are involved in trafficking and sorting of other GPCRs and signaling molecules through post-translational modifications, including ubiquitination. For example, ARRDC1 and ARRDC3 were reported to play roles in the degradation of the Notch receptor (Puca & Brou, 2014 [DOI](#)) and in the ubiquitination of ALG-2-interacting protein X (ALIX) (Dores, Lin, N, Mendez, & Trejo, 2015 [DOI](#)). ARRDC1 contains a PSPA motif, which binds the tumor susceptibility gene 101 (TSG101) protein, an essential component of an endosomal sorting complex. ARRDC1 also recruits E3 ligases, such as WW domain-containing E3 ubiquitin protein ligase2 (WWP2), inducing ubiquitination of itself and the subsequent release of ARRDC1-associated microvesicles (Nabhan, Hu, Oh, Cohen, & Lu, 2012 [DOI](#)). Another well-known α -arrestin, TXNIP, was first named as vitamin D3-upregulated protein 1 (VDUP1) after verification that its gene is a vitamin D3 target in cancer cells (K. S. Chen & DeLuca, 1994 [DOI](#); Qayyum, Haseeb, Kim, & Choi, 2021 [DOI](#)). Since then, TXNIP had been reported to directly interact with thioredoxin, which is an essential component of the cellular redox system, to inhibit its activity as an antioxidant (Junn

et al., 2000 [\[1\]](#); Nishiyama et al., 1999 [\[2\]](#); Patwari, Higgins, Chutkow, Yoshioka, & Lee, 2006 [\[3\]](#)). TXNIP was also reported to inhibit glucose uptake by inducing the internalization of glucose transporter 1 (GLUT1) through clathrin-mediated endocytosis and by indirectly reducing GLUT1 RNA levels (Wu et al., 2013 [\[4\]](#)). Although the TXNIP is known to be localized in both cytoplasm and nucleus, biological functions of TXNIP have been mostly explored in cytoplasm but remained poorly characterized in nucleus.

A few α -arrestins appear to have evolutionarily conserved functions in both human and invertebrates. For instance, the Hippo signaling pathway, which impacts a variety of cellular processes such as metabolism, development, and tumor progression (Mo, Park, & Guan, 2014 [\[5\]](#); Pei et al., 2015 [\[6\]](#); Schutte et al., 2014 [\[7\]](#); Y. Wang et al., 2010 [\[8\]](#); Zhi, Zhao, Zhou, Liu, & Chen, 2012 [\[9\]](#)), was shown to be regulated by α -arrestin in both *Drosophila* (Y. Kwon et al., 2013 [\[10\]](#)) and human cells (Xiao et al., 2018 [\[11\]](#)). In *Drosophila*, the protein Leash was identified as an α -arrestin and shown to down-regulate Yki by promoting its lysosomal degradation, leading to a restriction in growth (Y. Kwon et al., 2013 [\[10\]](#)). In human cells, ARRDC1 and ARRDC3 were shown to induce degradation of the mammalian homolog of Yki, YAP1, by recruiting the E3 ubiquitin ligase ITCH in renal cell carcinoma (Xiao et al., 2018 [\[11\]](#)), suggesting functional homology between human and *Drosophila*. However, because the α -arrestins interact with multiple targets, an unbiased, comparative analysis of interactome is required to determine whether other α -arrestin from human and *Drosophila* have common and specific interacting partners, which will determine their functional homology and diversification.

A comprehensive understanding of their protein-protein interactions (PPIs) and interactomes will shed light on the underlying molecular mechanisms, reveal novel regulatory axes, and enable the identification of previously unrecognized roles of α -arrestin in cellular processes. Furthermore, extensive characterization of the α -arrestin interactome may help uncover potential therapeutic targets and provide valuable insights into the treatment of diseases associated with dysregulated signaling pathways (Diaz et al., 2005 [\[12\]](#); Lu, Simin, Khan, & Mercurio, 2008 [\[13\]](#); Q. Y. Wang et al., 2018 [\[14\]](#); Zhou, Tardivel, Thorens, Choi, & Tschopp, 2010 [\[15\]](#)).

In this study, we conducted affinity purification/mass spectrometry (AP/MS) of six human and twelve *Drosophila* α -arrestins. A high-confidence PPI network was constructed by selecting a cut-off for receiver operating characteristic (ROC) curves of Significance Analysis of INteractome express (SAINTexpress) scores (Teo et al., 2014 [\[16\]](#)). The constructed interactomes were validated using known affinities between domains of prey proteins and the short linear motifs of α -arrestins. We also investigated orthologous relationships between binding partners from human and *Drosophila* and found that many proteins with both known and novel functions could be conserved between two species. Finally, we performed experiments to provide new insights into the functions of TXNIP and ARRDC5 that were revealed in our study. Together, our results provide a valuable resource that describes the PPI network for α -arrestins in both human and *Drosophila* and suggest novel regulatory axes of α -arrestins.

Results

High-confidence α -arrestin interactomes in human and *Drosophila*

Genome-scale sets of prey proteins interacting with α -arrestins (referred to herein as ‘interactomes’) were compiled by conducting AP/MS for six human and twelve *Drosophila* α -arrestin proteins (Figure 1A [\[17\]](#); Table S1). Proteins possibly interacting with α -arrestins were pulled down from total cell lysates of human embryonic kidney 293 (HEK293) and S2R+ cells stably expressing GFP-tagged α -arrestins (Figure 1B [\[18\]](#); Figure S1 [\[19\]](#)). All α -arrestin experiments

were replicated twice, and negative control experiments were conducted multiple times. In total, 3,243 and 2,889 prey proteins involved in 9,908 and 13,073 PPIs with human and *Drosophila* α -arrestins, respectively, were initially detected through AP/MS (**Figure 1B** [↗](#); Table S2).

To build high-confidence interactomes of α -arrestin family proteins, a probabilistic score for individual PPIs was estimated using SAINTexpress (Teo et al., 2014 [↗](#)) and an optimal cutoff for the scores was set using positive and negative PPIs of α -arrestin from public databases and the literature (Colland et al., 2004 [↗](#); Dotimas et al., 2016 [↗](#); Draheim et al., 2010 [↗](#); Mellacheruvu et al., 2013 [↗](#); Nabhan et al., 2012 [↗](#); Nishinaka et al., 2004 [↗](#); Puca & Brou, 2014 [↗](#); Szklarczyk et al., 2015 [↗](#); Warde-Farley et al., 2010 [↗](#); Wu et al., 2013 [↗](#)) (Tables S2). The resulting ROC curves showed high area under curve (AUC) values and the SAINTexpress scores at which the false discovery rate (FDR) was 0.01 were selected as cutoffs (0.85 for human and 0.88 for *Drosophila*, **Figure 1C** [↗](#)). Given the cutoffs, 1,306 and 1,732 PPIs involving 902 and 1,732 proteins were selected for human and *Drosophila*, respectively. Because proteins of low abundance (low spectral counts) are easily affected by a stochastic process (Lundgren, Hwang, Wu, & Han, 2010 [↗](#); Old et al., 2005 [↗](#)), the minimum spectral count of PPIs was set at 6, allowing us to select PPIs with higher confidence. In fact, the spectral counts of the filtered PPIs were highly reproducible between replicates (**Figure S2A** [↗](#); Pearson's correlations, 0.91 for human; 0.89 for *Drosophila*) and principal component analysis based on $\log_2$ spectral counts also confirmed a high reproducibility between replicates (**Figure S2B** [↗](#)). Moreover, we successfully detected many known interaction partners of α -arrestins such as NEDD4, WWP2, WWP1, ITCH and TSG101, which have been previously reported in the literature and PPI databases (**Figure S2C** [↗](#)) (Colland et al., 2004 [↗](#); Dotimas et al., 2016 [↗](#); Draheim et al., 2010 [↗](#); Mellacheruvu et al., 2013 [↗](#); Nabhan et al., 2012 [↗](#); Nishinaka et al., 2004 [↗](#); Puca & Brou, 2014 [↗](#); Szklarczyk et al., 2015 [↗](#); Warde-Farley et al., 2010 [↗](#); Wu et al., 2013 [↗](#)). Finally, our filtered interactomes of human and *Drosophila* α -arrestins, comprised of 390 PPIs between six α -arrestins and 307 prey proteins in human, and 740 PPIs between twelve α -arrestins and 467 prey proteins in *Drosophila*, are hereafter referred to as 'high-confidence PPIs' (Table S3).

Short-linear motifs and protein domains enriched in α -arrestins and their interactomes

To validate our high-confidence PPIs, we sought to analyze known short-linear motifs in α -arrestins, which are commonly 3-15 stretches of amino acids that are known to participate in interactions with other protein domains (Dinkel et al., 2015 [↗](#)). Utilizing the known affinities between short linear motifs in α -arrestins and protein domains in interactomes from eukaryotic linear motif (ELM) database (Dinkel et al., 2015 [↗](#)), we evaluated whether our high-confidence PPIs could be explained by the known affinities between them (Table S4). The fractions of our high-confidence PPIs (green, **Figure 1D** [↗](#) top), supported by the known affinities were significantly greater than those of all raw PPIs (red, **Figure 1D** [↗](#) top) in both species ($P < 9.37 \times 10^{-11}$ for human and $P < 0.0012$ for *Drosophila*, one-sided Fisher's exact test, **Figure 1D** [↗](#) top). One of the most well-known short-linear motifs in α -arrestin is PPxY, which is reported to bind with high affinity to the WW domain found in various proteins, including ubiquitin ligases (Macias et al., 1996 [↗](#)). Our analysis revealed the specific enrichment of WW domain-containing proteins in the interactomes of α -arrestins with at least one PPxY motif but not in that of the human α -arrestin (ARRDC5) without a PPxY motif (**Figure 1D** [↗](#), bottom-left). The interactomes of five out of the eight *Drosophila* α -arrestins with a PPxY motif were enriched for WW domain containing proteins, but there was no such enrichment for any of the *Drosophila* α -arrestins without a PPxY motif (**Figure 1D** [↗](#), bottom-right). In conclusion, a considerable portion of the high-confidence PPIs identified in this study can be evident by known affinities between short linear motifs and protein domains.

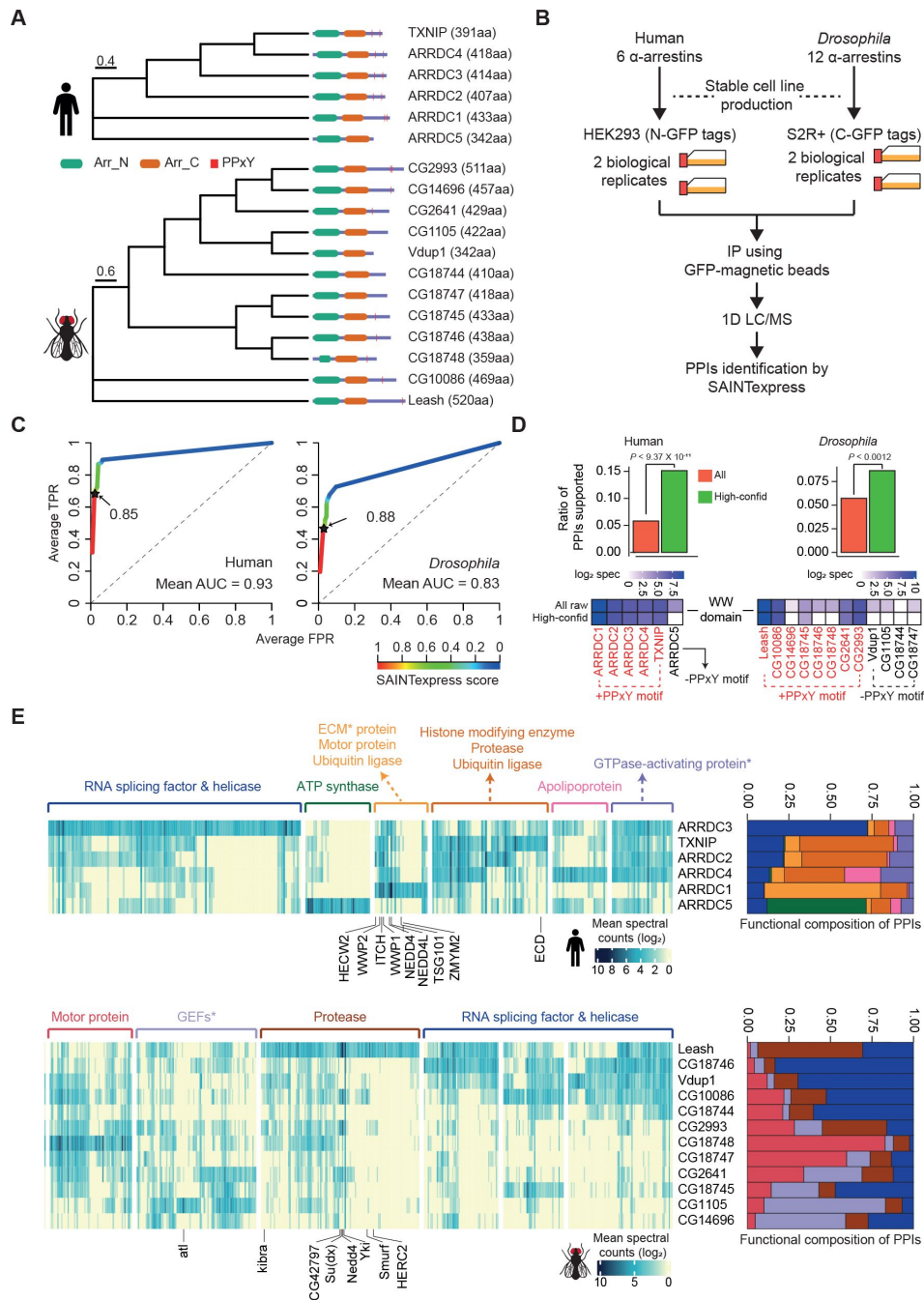


Figure 1.

Identification of high-confidence α-arrestin PPIs

(A) Phylogenetic tree of α-arrestins from human (6, top) and *Drosophila* (12, bottom) based on protein sequences. The numbers in parentheses indicate the length of each protein. aa, amino acids; Arr_N: Arrestin N domain; Arr_C: Arrestin C domain; PPxY: PPxY motif. **(B)** Shown is a schematic flow of AP/MS experiments and computational analysis. **(C)** ROC curves of SAINTExpress scores along with AUC values. The arrows point to the cutoff scores used in subsequent studies in human (left) and *Drosophila* (right). **(D)** (Top) The fraction of high-confidence and all raw (unfiltered) PPIs that are supported by known affinities between short linear motifs and protein domains in human (left) and *Drosophila* (right). One-sided, Fisher's exact test was performed to test the significance. (Bottom) The sum of log₂ spectral counts (log₂ spec) of interacting proteins with WW domains observed in the high confidence and all raw PPIs are visualized in the heatmap. **(E)** The α-arrestins and interacting prey proteins were hierarchically clustered based on the log₂ mean spectral counts and summarized for human (top) and *Drosophila* (bottom) in the heatmaps. The functionally enriched protein groups of preys are indicated at the top. Previously reported proteins interacting with α-arrestins are labeled at the bottom. On the right, the functional composition of prey groups is summarized with the sum of log₂ mean spectral counts of each prey group, which are colored to correspond with the labels on the left.

Next, we conducted enrichment analyses of proteins domains among interactome of each arrestin to investigate known and novel protein domains commonly or specifically interacting with α -arrestins (**Figure S3A** [↗](#); Table S5). The most prominent interacting domains in both species were the Homologous to E6AP C-terminus (HECT), WW, and C2 domains (**Figure S3A** [↗](#); Table S5). HECT and C3 domains are well known to be embedded in the E3 ubiquitin ligases such as NEDD4, HECW2, and ITCH along with WW domains (Weber, Polo, & Maspero, 2019 [↗](#)) and as we observed strong preference of WW domains to PPxY containing proteins (**Figure 1D** [↗](#)), these domains were significantly enriched in binding proteins of α -arrestins with PPxY motif in human and *Drosophila* (FDR < 0.033 ~ 1.23×10^{-11} for human; FDR < 0.045 ~ 4.10×10^{-6} for *Drosophila*, **Figure S3A** [↗](#); Table S5). Other common protein domains involved in the protein degradation process, such as proteasome domains, were also significantly enriched in the interactomes (of ARRDC4 in human and Leash in *Drosophila*) in both species (FDR < 6.41×10^{-4} for human and FDR < 1.30×10^{-5} for *Drosophila*, **Figure S3A** [↗](#); Table S5). Interestingly, some α -arrestins (ARRDC3 in human and Vdup1, Leash, and CG18746 in *Drosophila*) appeared to interact in common with RNA binding domains, such as DEAD box, helicase, WD40, and RNA recognition motif, but others did not. In addition, the cargo and motor protein domains IBN_N (FDR < 0.0076 for human and FDR < 2.50×10^{-4} ~ 2.11×10^{-6} for *Drosophila*) and myosin_head (FDR < 0.033 for human and FDR < 2.11×10^{-6} for *Drosophila*) also interacted with several α -arrestins in common (ARRDC4 in human and CG1105, CG18745, and CG18748 in *Drosophila*, **Figure S3A** [↗](#); Table S5). These enriched domains explain the conserved interactomes associated with RNA splicing and protein transport in both species. In addition, human α -arrestins seem to interact with human-specific domains, such as PDZ, Rho-GEF, MCM, laminin, zinc finger, and BAG6 domains, providing an expanded interactome of human α -arrestins (**Figure S3A** [↗](#); domains in black), indicating the presence of both conserved and specific protein domains interacting with α -arrestins.

Expanded functional signatures of α -arrestin interactomes

Because the functions of α -arrestins can be inferred based on their binding partners, the prey proteins were grouped based on their interactions with α -arrestins, which revealed specialized functions of the respective α -arrestins with some redundancy as well as both known and novel functions (**Figure 1E** [↗](#)). The analysis of protein class enrichment by the PANTHER classification system (Thomas et al., 2003 [↗](#)) revealed previously reported functions, such as ‘Ubiquitin ligase’ (FDR < 0.0019 and 5.01×10^{-7} for human; Benjamini-Hochberg correction) and ‘Protease’ (FDR < 1.93×10^{-6} for human and 5.02×10^{-6} for *Drosophila*) (Dores et al., 2015 [↗](#); Y. Kwon et al., 2013 [↗](#); Nabhan et al., 2012 [↗](#); Puca, Chastagner, Meas-Yedid, Israel, & Brou, 2013 [↗](#); Rauch & Martin-Serrano, 2011 [↗](#); Shea et al., 2012 [↗](#); Xiao et al., 2018 [↗](#)). In fact, the known binding partners, NEDD4, WWP2, WWP1, and ITCH in human and CG42797, Su(dx), Nedd4, Yki, Smurf, and HERC2 in *Drosophila*, are related to ubiquitin ligases and protein degradation (**Figure 1E** [↗](#); **Figure S2C** [↗](#)). In addition, novel biological functions of α -arrestins were uncovered. For instance, in human, prey proteins interacting with ARRDC3 displayed enrichment of ‘RNA splicing factor and helicase’ functions as well as ‘GTPase-activating proteins’, those of ARRDC4 were enriched with ‘Apolipoprotein’, and those of ARRDC5 with ‘ATP synthase’ (**Figure 1E** [↗](#), up). Motor protein, protease, ubiquitin ligase, RNA splicing factor, and helicase were functions that were also enriched in *Drosophila* prey proteins (**Figure 1E** [↗](#), bottom). Among them, the motor protein and RNA splicing, and helicase functions seemed to be novel conserved functions between human and *Drosophila*. The functional compositions of the interacting proteins summarized the common or highly specialized functions of α -arrestins well (**Figure 1E** [↗](#), right panel). For example, in human, proteins that interacted with TXNIP, ARRDC2, and ARRDC4 showed similar ubiquitination and protease-related functions, whereas ARRDC3 and ARRDC5 displayed unique interactomes associated with other functions. For *Drosophila*, the interactomes of the [Vdup1, CG10086 and CG18744], [CG18748 and CG18747], and [CG1105 and CG14696] α -arrestin subsets each exhibited similar functional compositions, but the Leash interactome showed a distinct enrichment of ubiquitination-related and protease functions. Taken together, these results suggest that the resulting high-confidence PPIs of α -arrestins expanded the functional interactome maps of α -arrestins in both human and *Drosophila*.

Subcellular localizations of α -arrestin interactomes

Cellular localizations of proteins often provide valuable information of their functions and activity, but only a small number of α -arrestins are known for their preferential subcellular localization. We thus examined the subcellular localizations of the interacting proteins using the cellular component feature in Gene Ontology (GO) using DAVID (Huang da, Sherman, & Lempicki, 2009a [↗](#), 2009b) (**Figure S3B** [↗](#); Table S6). Prey proteins (246 for human and 245 for *Drosophila*) that were localized in at least one cellular compartment were examined. We found that prey proteins of ARRDC5 were preferentially localized in the endoplasmic reticulum and at the plasma membrane (PM) but were less often localized in the nucleus, compared to those of other human α -arrestins (**Figure S3B** [↗](#), left). Similarly, the prey proteins of ARRDC1 and 4 were less often localized in the nucleus, instead being preferentially localized in the cytoplasm (ARRDC4) or extracellular space (ARRDC1), in agreement with previous reports (Nabhan et al., 2012 [↗](#); Q. Y. Wang et al., 2018 [↗](#)). TXNIP seemed to preferentially interact with prey proteins in cytoplasm and nucleus (**Figure S3B** [↗](#), left), consistent with a previous report (S. K. Kim, Choe, & Park, 2019 [↗](#); Saxena, Chen, & Shalev, 2010 [↗](#)). ARRDC3, which was suggested to be localized in cytoplasm in previous study (Nabhan et al., 2010 [↗](#)), appeared to interact with proteins preferentially localized in nucleus in addition to the ones in cytoplasm, implying novel functions of ARRDC3 in the nucleus. In *Drosophila*, the localization of interacting proteins is often uncharacterized compared to human, but a preference for a localization for part of the interactomes can be observed (**Figure S3B** [↗](#), right). Some of them are preferentially localized at the PM (CG18747), mitochondria (CG14696), peroxisome (CG14696), lysosome (CG2641), or cytoskeleton (CG18748), compared to others. However, interactomes of Leash, Vdup1, CG2641, CG18745, CG18746, and CG10086 are preferentially localized in the nucleus. Taken together, these data about the preferential localizations of interacting proteins provide evidence about the functions and activity of α -arrestins in cells.

Functional complexes in α -arrestin interactomes

The fact that protein functions are often realized in complexes (Hartwell, Hopfield, Leibler, & Murray, 1999 [↗](#)) urged us to search for functional complexes that extensively interact with α -arrestins. For this analysis, protein complexes that are significantly connected with each α -arrestin were examined using the COMpLex Enrichment Analysis Tool (COMPLEAT) (Vinayagam et al., 2013 [↗](#)), resulting in the detection of 99 and 18 protein complexes for human and *Drosophila*, respectively (Table S7). The complexes were iteratively combined with cellular components from GO (Table S7) based on the overlap coefficients (Vijaymeena & Kavitha, 2016 [↗](#)). The significance of the resulting combined complexes was then tested with the connectivity to each α -arrestin using the interquartile means (IQMs) of SAINTexpress scores compared to those from 1000 random cohorts ($P < 0.05$). This approach showed that 44 clustered complexes comprising 324 protein subunits were significantly interacting with six human α -arrestins (**Figure 2A** [↗](#); Table S7) and 21 clustered complexes comprising 192 subunits were significantly interacting with *Drosophila* α -arrestins (**Figure 2B** [↗](#); Table S7).

The two largest complexes found to interact with α -arrestins were related to protein degradation (proteasome and ubiquitin-dependent proteolysis) and RNA splicing and processing in both species (**Figure 2** [↗](#); Table S7). ARRDC1, 2, and 4 and TXNIP in human and Leash and CG2993 in *Drosophila* were found to interact with protein degradation complexes. CG2993 and CG18747 appeared to bind to a putative complex comprising NEDD4 family interacting protein 2, recently reported to be a mediator of ubiquitin ligase (Trimper et al., 2017 [↗](#)). On the other hand, ARRDC2, 3, and 4 in human and Leash, CG18746, Vdup1, CG10086, and CG18744 in *Drosophila* were found to interact with RNA splicing and processing complexes. Although the above-mentioned α -arrestins interacted in common with the two complexes described above, they were also found to bind to distinct complexes. For instance, TXNIP specifically binds to transcriptional and histone deacetylase (HDAC) complexes, ARRDC1 to axon guidance, endosomal sorting, and laminin complexes, ARRDC2 to the Set1C/COMPASS complex, ARRDC3 to transcription elongation factors

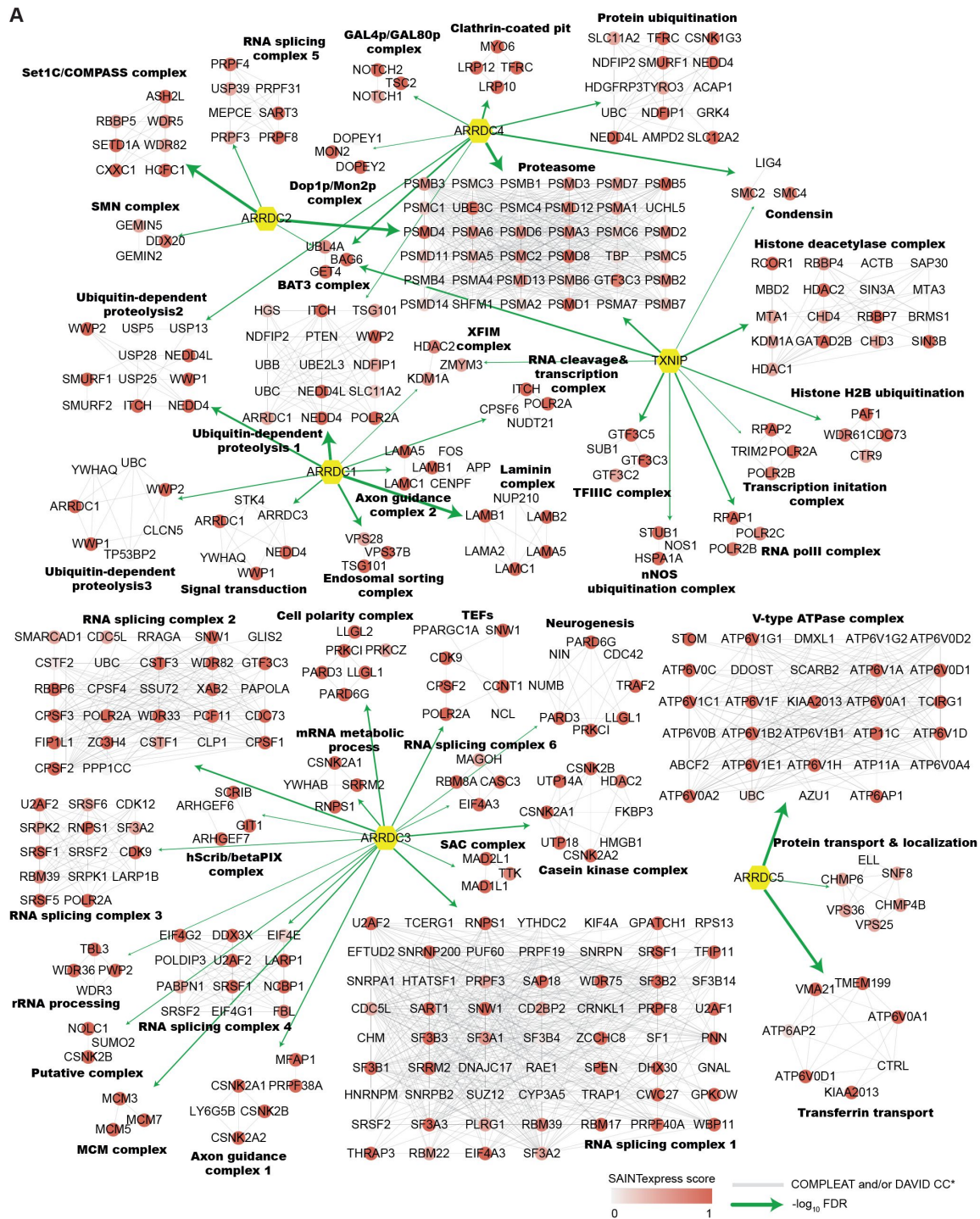


Figure 2.

Network of α -arrestins and their interacting protein complexes

Network of α -arrestins and the functional protein complexes that significantly interact with them in human (A) and *Drosophila* (B). α -arrestins are colored yellow and prey proteins in protein complexes are colored according to the SAINTexpress scores of the PPIs. The gray edges indicate that evidence supporting the complex was provided by COMPLEAT and/or GO cellular components. The thickness of the green arrows indicates the strength of the interaction between α -arrestins and the indicated protein complexes, which was estimated with $-\log_{10}$ FDR of complex association scores.

COMPASS, complex proteins associated with Set1; SMN, survivor of motor neurons; TFIIC, transcription factor III C; RNA polII, RNA polymerase II; MCM, minichromosome maintenance protein complex; SAC, spindle assembly checkpoint; NSL, non-specific lethal; WASH, Wiskott-Aldrich syndrome protein and scar homolog; Arp2/3, actin related protein 2/3. TEF, transcription elongation factor.

B

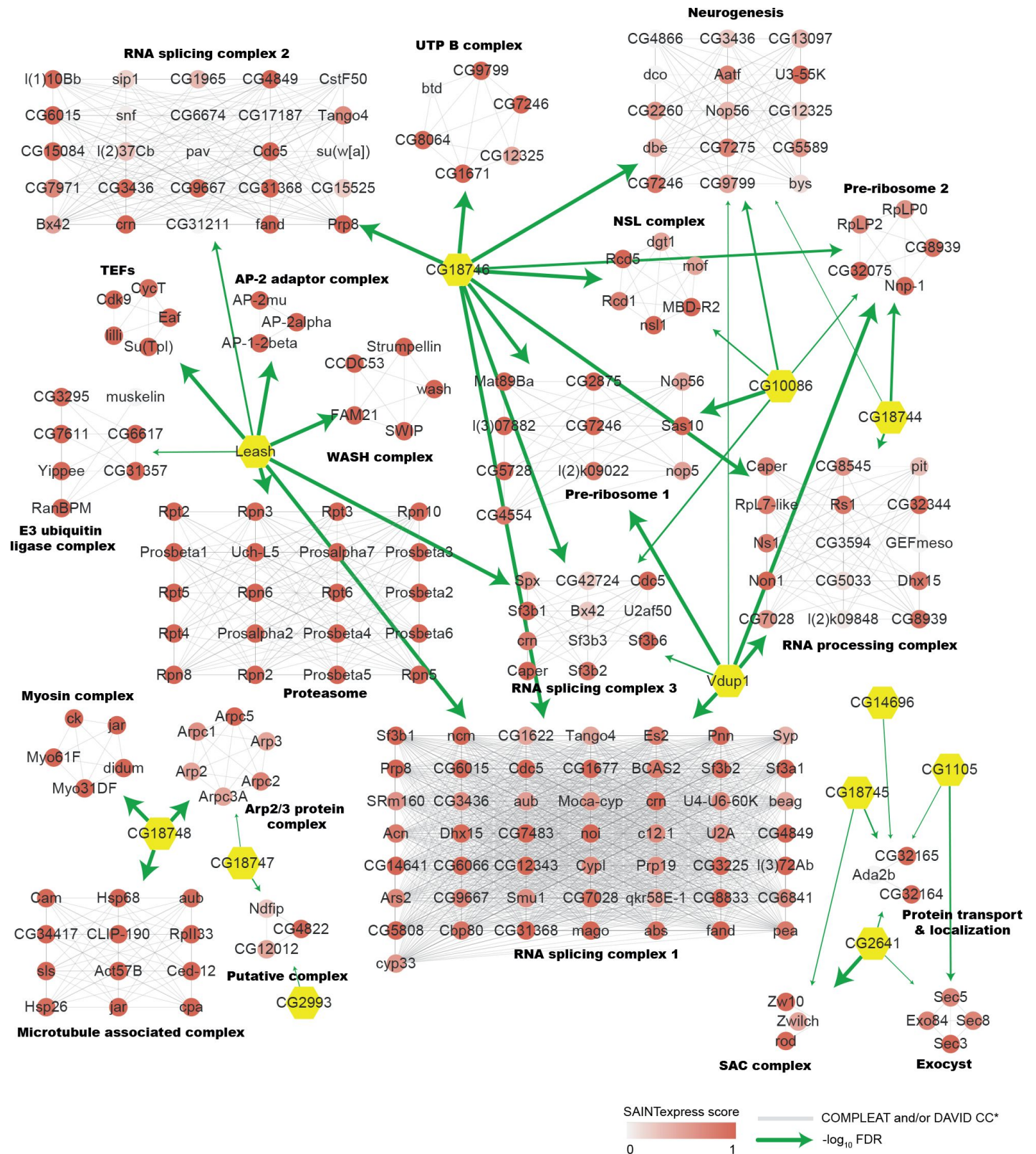


Figure 2. (continued)

and spindle assembly checkpoint and cell polarity complexes, and ARRDC4 to clathrin-coated pit and BAT3 complexes in human. In *Drosophila*, Leash specifically binds to AP-2 adaptor and WASH complexes and CG18746 to the UTP B complex. In addition to the two largest complexes and their associated α -arrestins, ARRDC5 in human and CG2641, CG1105, CG14696, and CG18745 in *Drosophila* interact in common with protein transport and localization complexes. ARRDC5 is specifically associated with V-type ATPase and vacuolar protein sorting complexes in human. CG18748 and CG18747 are associated with motor protein complexes including actin, myosin, and microtubule-associated complexes in *Drosophila*. Taken together, the results from this analysis provide a glimpse of underexplored roles for α -arrestins in diverse cellular processes.

Conserved interactomes of α -arrestins

Given that α -arrestins are widely conserved in metazoans (DeWire, Ahn, Lefkowitz, & Shenoy, 2007), we sought to exploit the evolutionally conserved interactomes of human and *Drosophila* α -arrestins. For this analysis, we searched for orthologous relationships in the α -arrestin interactomes using the DRSC Integrative Ortholog Prediction Tool (DIOPT) (Hu et al., 2011). Among high-confidence prey proteins, 68 in human and 64 in *Drosophila* were reciprocally predicted to have ortholog relationships, defining 58 orthologous prey groups (DIOPT score ≥ 2 ; Table S8). α -arrestins were then hierarchically clustered based on the $\log_2$ transformed mean spectral counts of these orthologous interactome, defining seven groups of α -arrestins. Orthologous prey proteins were grouped according to their shared biological function, defining nine functional groups and others of diverse functions (Figure 4). The resulting clusters revealed PPIs that were functionally conserved. For instance, ARRDC3 in human and CG18746 in *Drosophila* actively interact with proteins in RNA binding and splicing groups. Leash in *Drosophila* appeared to interact with proteins in similar functional groups as ARRDC3 but, like ARRDC1, it also extensively interacts with members of ubiquitin-dependent proteolysis groups. In addition, ARRDC4 interacts with proteins in the motor protein and trafficking group, similar to CG18748 in *Drosophila*, and binds to proteins in the ubiquitin dependent proteolysis group, similar to TXNIP. Similarly, CG10086 and Vdup1, CG14696 and ARRDC5, and CG2993 and ARRDC2 appeared to have conserved interactomes between human and *Drosophila*.

The most prominent functional modules shared across both species were the ubiquitin dependent proteolysis, endosomal trafficking, and small GTPase binding modules, which are in agreement with the well-described functions of α -arrestins in membrane receptor degradation through ubiquitination and vesicle trafficking (Dores et al., 2015; Nabhan et al., 2012; Puca et al., 2013; Xiao et al., 2018) (Figure 4). In contrast, the functional modules involving cyclin and cyclin-dependent kinase, casein kinase complex, and laminin seemed to be conserved between relatively specific sets of α -arrestins, whereas those related to motor proteins and RNA binding and splicing were more generally conserved. Taken together, the comparative analyses led us to identification of detailed, orthologous interactome maps of α -arrestins, which extend beyond the limited insights provided by sequence-based comparative analysis alone (Figure S4). Conserved roles of α -arrestins in both established and previously uncharacterized signaling pathways expand our understanding of the diverse roles of α -arrestins in cellular signaling.

Chromatin accessibility is globally decreased by TXNIP knock down

TXNIP is one of the most well-studied α -arrestins. Previous studies reported that TXNIP interacts with transcriptional repressors, such as FAZF, PLZF, and HDAC1 or HDAC3, to exert antitumor activity (S. H. Han et al., 2003) or repress NF- κ B activation (H. J. Kwon et al., 2010). However, although such studies provided information about interactions with a few transcriptional repressors, they barely provided a systematic view of the roles of TXNIP in controlling the chromatin landscape and gene expression. In that sense, our PPI analysis first revealed that TXNIP extensively binds to chromatin remodeling complexes, such as the HDAC and histone H2B ubiquitination complexes, as well as to transcriptional complexes, such as the RNA polymerase II and transcription factor IIIc complexes (Figure 2A). Such PPIs indicate that TXNIP could control

transcriptional and epigenetic regulators. To examine how the global epigenetic landscape is remodeled by TXNIP, we knocked down its expression in HeLa cells with a small interfering RNA (siTXNIP) and confirmed a decrease at both the RNA and protein levels (**Figure 4A** and **B**). We then produced two biological replicates of ATAC- and RNA-seq experiments in HeLa cells with TXNIP depletion (Table S9) to detect differentially accessible chromatin regions (dACRs) and differentially expressed genes (DEGs) (**Figure 4C**; Table S10). The replicated samples were well grouped by the siTXNIP condition in principle component spaces (**Figure S5A** and **B**). The normalized ATAC-seq signal and the RNA level of expressed genes clearly showed the enrichment of open chromatin signals around the transcription start sites (TSSs) of genes that are actively transcribed (**Figure S5C**). We detected 70,746 high confidence accessible chromatin regions (ACRs) across all samples, most of which were located in gene bodies (38.74%), followed by intergenic regions (32.03%) and promoter regions (29.23%, **Figure S6A**). TXNIP knockdown appeared to induce a global decrease in chromatin accessibility in many genomic regions including promoters (**Figure 4D**). Of the high confidence ACRs, 7.38% were dACRs under TXNIP depletion; most dACRs showed reduced chromatin accessibility under this condition (dACRs(-), **Figure 4D**; **Figure S6B**). dACRs(-) were preferentially localized in gene bodies, whereas dACRs(+) were more often observed in promoter regions (**Figure S6C**).

The global chromatin changes induced by TXNIP knockdown could impact gene expression at corresponding loci. In fact, our gene expression analysis showed that 956 genes were downregulated, and 295 genes were upregulated by TXNIP knockdown compared to the control (**Figure 4E**), suggesting that the global decrease in chromatin accessibility induced by TXNIP depletion would mediate the repression of gene expression. To confirm this phenomenon, we first selected sets of differentially (“Down genes” and “Up genes” in **Figure 4E**, “Down” and “Up” in **Figure 4F**) and non-differentially expressed genes (“None” in **Figure 4F**) with at least one detectable ACR in promoter or gene body. Next, the cumulative distribution function (CDF) of accessibility changes demonstrated that the genes with a decreased RNA level (“Down”) showed significantly reduced chromatin accessibilities at promoters compared to those with no changes in the RNA level (“None”) (**Figure 4F**; $P < 5.81 \times 10^{-28}$ for max changes; **Figure S6D**; $P < 3.76 \times 10^{-32}$ for mean changes, Kolmogorov-Smirnov (KS) test). In contrast, genes with increased RNA expression (“Up”) exhibited no changes in chromatin accessibility at the promoter (**Figure 4F**; $P < 0.68$ for max changes; **Figure S6D**; $P < 0.49$ for mean changes, KS test), indicating that chromatin opening at promoters is necessary but not sufficient to induce gene expression. ACRs located in gene bodies also showed a similar trend: genes with a decreased RNA level (“Down”) showing decreased chromatin accessibility upon TXNIP depletion (**Figure S6E**; $P < 0.002$ for max changes and $P < 7.68 \times 10^{-7}$ for mean changes, KS test), suggesting that TXNIP is likely to be a negative regulator of chromatin repressors that induce heterochromatin formation. We then used GO analysis (Raudvere et al., 2019) to examine the biological functions of genes that exhibited decreased chromatin accessibility at their promoter and decreased RNA expression upon TXNIP knockdown (Table S11). In general, genes associated with developmental process, signaling receptor binding, cell adhesion and migration, immune response and extracellular matrix constituents appeared to be repressed upon TXNIP depletion (**Figure 4G**).

TXNIP represses the recruitment of HDAC2 to target loci

Given that TXNIP knockdown led to a global reduction in chromatin accessibility with decreased transcription, we focused on identifying the potential role of the epigenetic silencer HDAC2, one of the strong binding partners of TXNIP in the AP/MS analysis, in mediating the TXNIP-dependent epigenetic and transcriptional modulation. Consistent with the AP/MS data, immunoprecipitation (IP) experiments showed that the two proteins indeed interact with each other. Furthermore, TXNIP knockdown reduced the amount of TXNIP-interacting HDAC2 protein but did not affect the HDAC2 expression level (**Figure 5A**). To find out how the TXNIP HDAC2 interaction impacts the epigenetic and transcriptional reprogramming of target loci, we first checked whether the TXNIP-HDAC2 interaction causes cytosolic retention of HDAC2 to inhibit nuclear HDAC2-mediated global

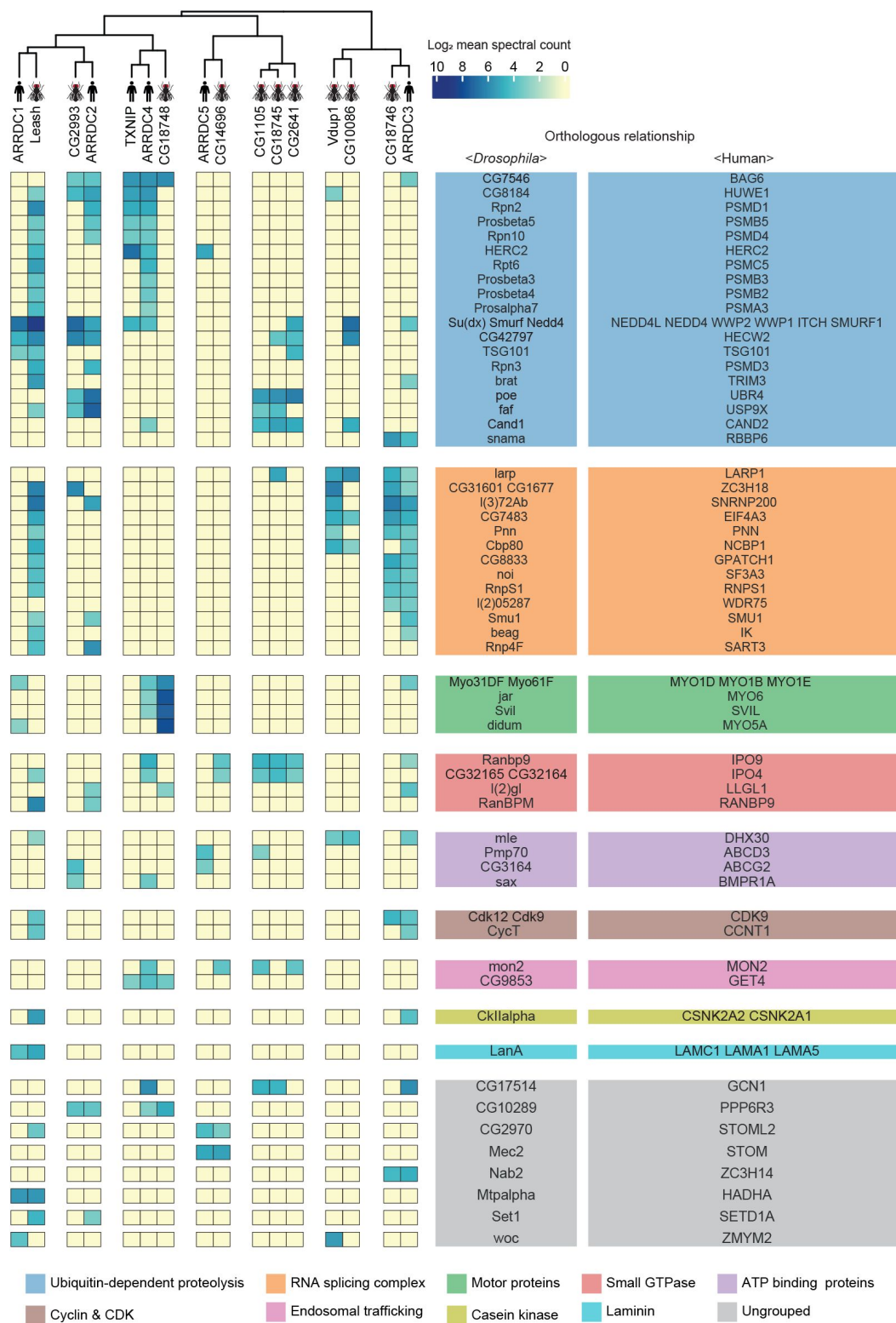


Figure 3.

A substantial fraction of α -arrestin-PPIs are conserved across species

Human and *Drosophila* α -arrestins are hierarchically clustered based on log₂-transformed mean spectral counts of their orthologous interactome. They are then manually grouped according to shared biological functions and assigned distinct colors. The names of orthologous proteins that interact with α -arrestins are displayed on the right side of the heatmap.

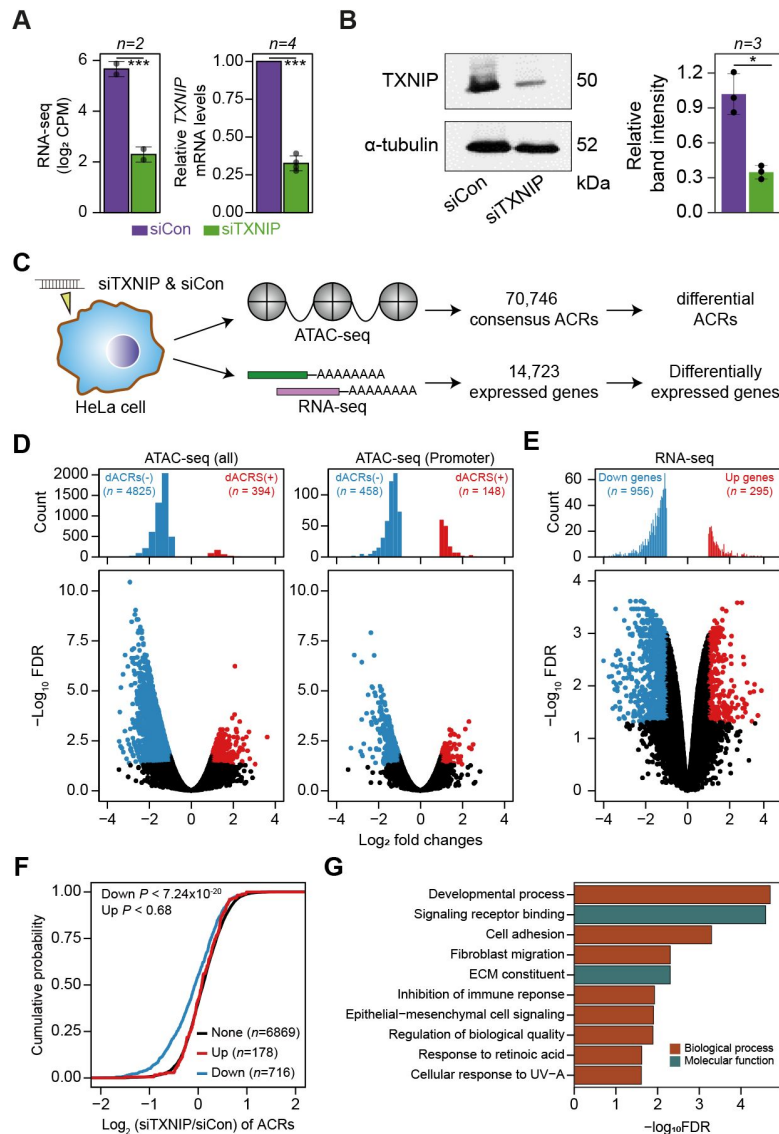


Figure 4.

TXNIP knockdown induces a global decrease in chromatin accessibility and gene expression

(A-B) HeLa cells were treated with either siRNA against TXNIP (siTXNIP) or negative control (siCon) for 48 hours (hr) and analyzed of changes in the mRNA **(A)** and protein levels **(B)** of TXNIP. **(A)** Expression levels of RNAs were quantified by RNA-seq (left, log₂ counts per million mapped reads (CPM), see “Processing of RNA-seq data” in “Materials and Methods”) and RT-qPCR (right, relative levels of TXNIP in siTXNIP compared to siCon condition, see “Quantitative Reverse-transcription PCR” in Supplementary Information). **(B)** Protein levels were first visualized by western blot analysis of lysates from HeLa Cells and band intensities of three independent experiments were quantified (right). **(A-B)** Gray dots depict actual values of each experiment and bar plots indicate mean $\pm$ standard deviation (sd). ***FDR < 0.001 (test of differential expression by edgeR (Robinson et al., 2010), see “Processing of RNA-seq data” in “Materials and Methods”) for RNA-seq. *P < 0.05, ***P < 0.001 (two-sided paired Student T test) for RT-qPCR and western blots. **(C)** A schematic workflow for detecting dACRs and DEGs using ATAC- and RNA-seq analyses, respectively. **(D)** Volcano plots of differential chromatin accessibility for all ACRs (left) and those associated with promoters (right). **(E)** Volcano plots of differential gene expression. **(D-E)** Blue dots denote “dACRs(-)” of significantly decreased chromatin accessibility **(D)** and “Down” genes of significantly down regulated genes **(E)** in siTXNIP-treated cells (FDR $\leq$ 0.05, log₂(siTXNIP / siCon) $\leq$ -1); red dots denote “dACRs(+)” of significantly increased chromatin accessibility **(D)** and “Up” genes of significantly up-regulated genes **(E)** in siTXNIP-treated cells (FDR $\leq$ 0.05, log₂(siTXNIP / siCon) $\geq$ 1). Black dots denote data points with no significant changes. **(F)** Changes in chromatin accessibility of ACRs located in the promoter region of genes were plotted as CDFs. Genes were categorized into three groups based on changes in RNA levels (“Up”, “Down” as in **(E)**) and “None” indicating genes with $-0.5 \leq \log_2(\text{siTXNIP} / \text{siCon}) \leq 0.5$. The number of genes in each group are shown in parentheses and P values in the left upper corner were calculated by one-sided KS test. **(G)** Top 10 GO terms (biological process and molecular function) enriched in genes that exhibited decreased chromatin accessibility at their promoter and decreased RNA expression upon TXNIP knockdown (Table S11).

histone deacetylation. However, both the expression level and subcellular localization of HDAC2 were unaffected by a reduction in TXNIP, as confirmed by Western blot analysis using cytoplasmic and nuclear fractions as well as by an immunofluorescence assay (**Figure 5B** [↗](#) and **Figure 57A** [↗](#)), indicating that TXNIP might modulate HDAC2 activity in a different way.

We next asked if the transcriptional suppression of TXNIP-target genes was mediated by changes in HDAC2 recruitment to and histone acetylation of chromatin. To address this question, genes that were significantly downregulated by TXNIP knockdown and that contained at least one dACR in the promoter were selected by the following additional criteria: 1) the RNA level in normal HeLa cells is ≥ 10 TPM and 2) the total ATAC-seq read count at the promoter in siTXNIP-treated HeLa cells is reduced ≥ 1.5 -fold compared to that in normal cells. Among the four TXNIP-target genes selected by the above-mentioned criteria, the expression levels of CD22 and L1CAM were significantly reduced ($P < 0.05$, Student's T test, **Figure 57B** [↗](#)). The two genes were further examined to determine whether the levels of HDAC2 binding signal and histone acetylation in their promoter regions were changed upon TXNIP knockdown (**Figure 5C** [↗](#)). We observed that RNA- and ATAC-seq coverages in exonic and promoter region of CD22 and L1CAM genes were clearly reduced upon TXNIP depletion (**Figure 5C** [↗](#) top) and an analysis of chromatin immunoprecipitation (ChIP) signals for HDAC2 and histone H3 acetylation at each dACR(-) detected in the L1CAM and CD22 promoters revealed that TXNIP knockdown increased the recruitment of HDAC2 to TXNIP-target loci, accompanied by decreased histone H3 acetylation (**Figure 5C** [↗](#) bottom). Therefore, these results suggest that the TXNIP interaction with HDAC2 inhibits the chromatin occupancy of HDAC2 and subsequently reduces histone deacetylation to facilitate global chromatin accessibility.

ARRDC5 plays a role in osteoclast differentiation and function

Given that various subunits of the V-type ATPase interact with ARRDC5, we speculated that ARRDC5 might be involved in the function of this complex (**Figure 6A** [↗](#)). V-type ATPase plays an important role in the differentiation and function of osteoclasts, which are multinucleated cells responsible for bone resorption in mammals (Feng et al., 2009 [↗](#); Qin et al., 2012 [↗](#)). Therefore, we hypothesized that ARRDC5 might be also important for osteoclast differentiation and function. To determine whether ARRDC5 affects osteoclast function, we prepared osteoclasts by infecting bone marrow-derived macrophages (BMMs) with lentivirus expressing either GFP-GFP or GFP-ARRDC5 and differentiating the cells into mature osteoclasts. After five days of differentiation, ectopic expression of GFP-ARRDC5 had significantly increased the total number of tartrate resistant acid phosphatase (TRAP)-positive multinucleated cells compared to GFP-GFP overexpression (**Figure 6B** [↗](#)). In particular, the number of TRAP-positive osteoclasts with a diameter larger than 200 μm was significantly increased by GFP-ARRDC5 overexpression (**Figure 6B** [↗](#)), suggesting that ARRDC5 expression increased osteoclast differentiation. Additionally, the area of resorption pits produced by GFP ARRDC5-expressing osteoclasts in a bone resorption pit assay was approximately 4-fold greater than that of GFP-GFP expressing osteoclasts (**Figure 6C** [↗](#)). These results imply that the ectopic expression of ARRDC5 promotes osteoclast differentiation and bone resorption activity.

The V-type ATPase is localized at the osteoclast PM (Toyomura et al., 2003 [↗](#)) and its localization is disrupted by bafilomycin A1, which is shown to attenuate transport of the V-type ATPase to the membrane (Matsumoto & Nakanishi-Matsui, 2019 [↗](#)). We analyzed changes in V-type ATPase localization in GFP-GFP and GFP-ARRDC5 overexpressing osteoclasts. GFP signals were detected at the cell cortex when GFP-ARRDC5 was overexpressed, indicating that ARRDC5 might also localized to the osteoclast PM (**Figure 6D** [↗](#)). In addition, we detected more V-type ATPase signals at the cell cortex in the GFP-ARRDC5 overexpressing osteoclasts, and ARRDC5 and V-type ATPase were co-localized at the osteoclast membrane (**Figure 6D** [↗](#)). Notably, bafilomycin A1 treatment reduced not only the V-type ATPase signals detected at the cortex but also the GFP-ARRDC5 signals (**Figure 6D** [↗](#)). These results indicate that ARRDC5 might control the membrane localization of the V-type during osteoclast differentiation and function.

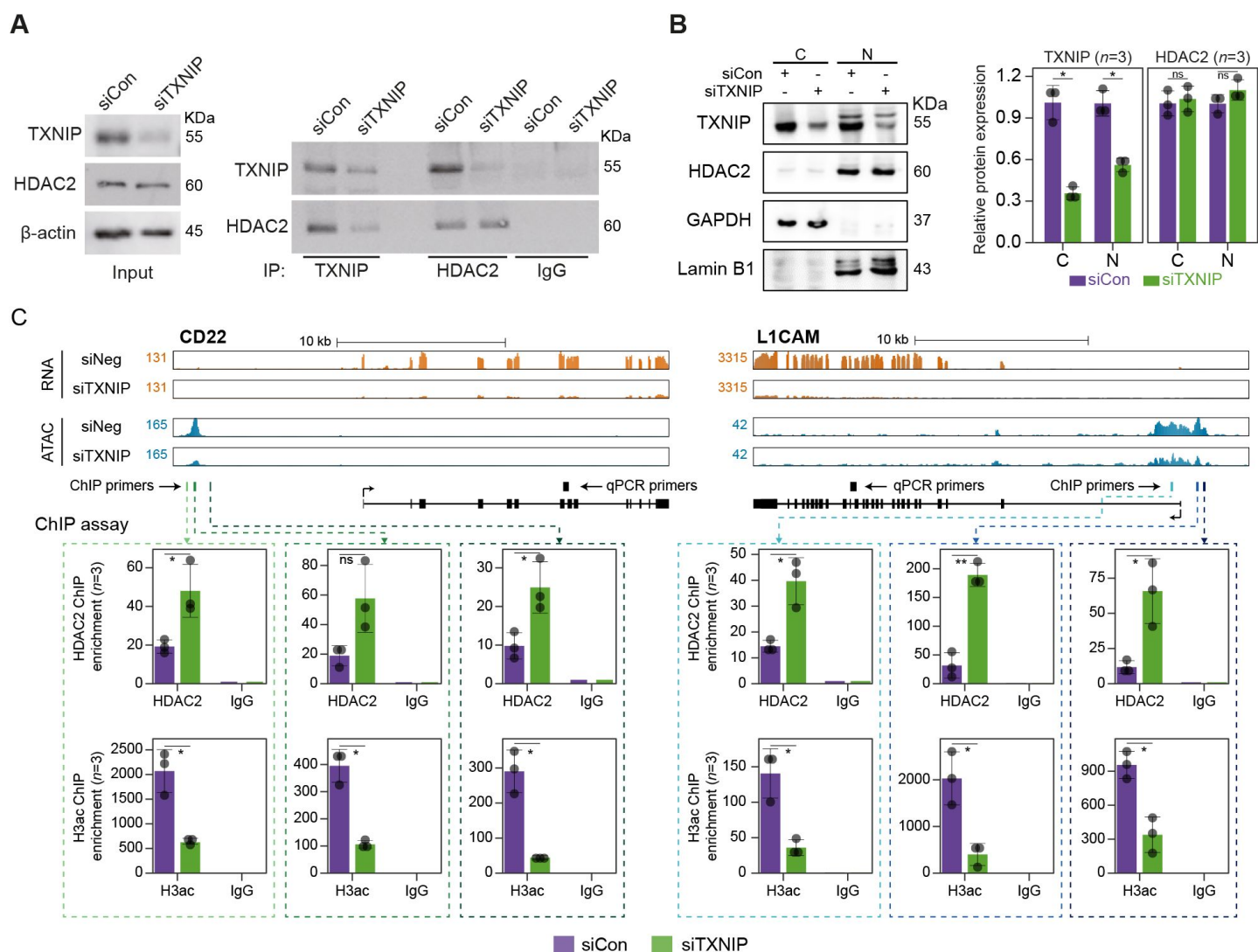


Figure 5.

TXNIP directly represses the recruitment of HDAC2 to target loci

(A) Analysis of co-IP between the TXNIP and HDAC2 proteins. Lysates from HeLa cells that had been treated with either siCon or siTXNIP for 48 hr were subjected to IP and immunoblotting with antibodies recognizing TXNIP and HDAC2, with IgG used as the negative control. **(B)** Nuclear and cytoplasmic fractions of HeLa cells were analyzed with Western blots following transfection with siCon or siTXNIP for 48 hr (left). Lamin B1 and GAPDH were used as nuclear and cytoplasmic markers, respectively. Western blot results from three independent experiments for TXNIP and HDAC2 were quantified as in **Figure 4B**. C, cytoplasm; N, nucleus. **(C)** Genomic regions showing RNA expression and chromatin accessibility at CD22 and L1CAM gene loci (top). Through the ChIP-qPCR analysis, the fold enrichment of HDAC2 and histone H3 acetylation (H3ac) at the CD22 and L1CAM promoter regions in HeLa cells treated with either siCon or siTXNIP for 48 hr were quantified (bottom). Data are presented as the mean $\pm$ sd ($n=3$, biological replicates). Gray dots depict actual values of each experiment. * $P < 0.05$, ** $P < 0.01$, ns: not significant (two-sided paired Student T test).

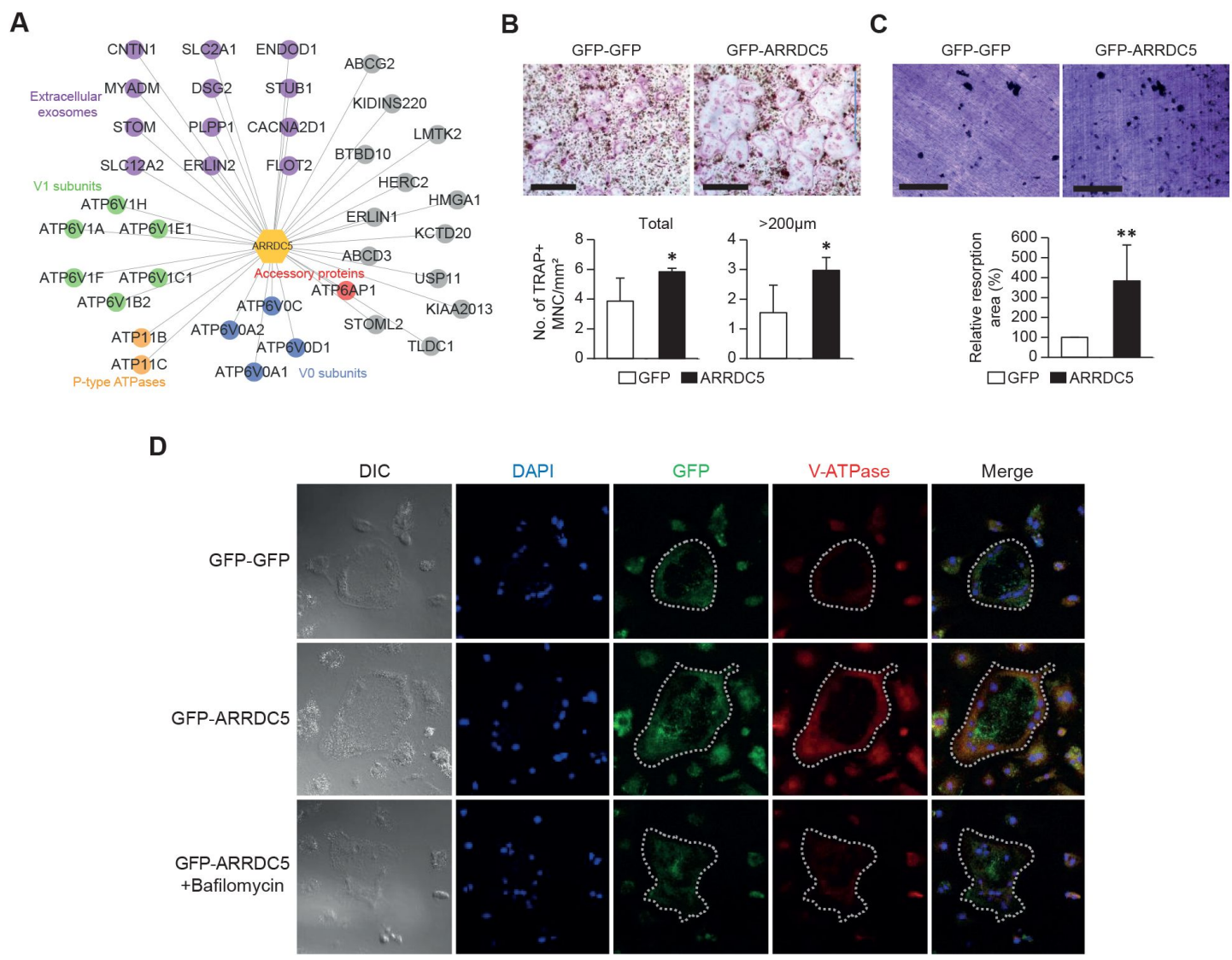


Figure 6.

Interaction of ARRDC5 with the V-type ATPase in osteoclasts

(A) The human ARRDC5-centric PPI network. V-type and P-type ATPases, their related components, and extracellular exosomes are labeled and colored. Other interacting proteins are indicated with gray circles. **(B)** TRAP staining of osteoclasts. Cell differentiation was visualized with TRAP staining of GFP-GFP or GFP-ARRDC5 overexpressing osteoclasts (scale bar = 500 μ m). TRAP-positive multinucleated cells (TRAP+ MNC) were quantified as the total number of cells and the number of cells whose diameters were greater than 200 μ m. * $P < 0.05$. **(C)** Resorption pit formation on dentin slices. Cell activity was determined by measuring the level of resorption pit formation in GFP-GFP or GFP-ARRDC5 overexpressing osteoclasts (scale bar = 200 μ m). Resorption pits were quantified as the percentage of resorbed bone area per the total dentin disc area using ImageJ software. The resorption area is relative to that in dentin discs seeded with GFP-GFP overexpressing osteoclasts, which was set to 100%. ** $P < 0.01$. **(D)** Localization of Arrdc5 and the V-type ATPase in osteoclasts. The V-type ATPase was visualized with immunofluorescence (red), GFP-GFP and GFP-ARRDC5 were visualized with GFP fluorescence (green), and nuclei were visualized with DAPI (blue). Representative fluorescence images are shown. Dashed lines were used to outline representative osteoclasts (scale bar = 100 μ m).

We constructed high-confidence interactomes of α -arrestins from human and *Drosophila*, comprising 307 and 467 interacting proteins, respectively. The resulting interactomes greatly expanded previously known PPIs involving α -arrestins and the majority of interactomes were first reported in this study, which needs to be validated experimentally (Tian, Kang, & Benovic, 2014 [DOI](#); Zbieralski & Wawrzycka, 2022 [DOI](#)). However, some known PPIs were missed in our interactomes due to low spectral counts and SAINTexpress scores, probably resulting from different cellular contexts, experimental conditions, or other factors (Figure S2C [DOI](#)).

Integrative map of protein complexes that interact with α -arrestins (Figure 2 [DOI](#)) hint towards many aspects of α -arrestins's biology that remain uncharacterized. For example, role of α -arrestins in the regulation of β 2AR in human remained controversial. One study proposed that α -arrestins might act coordinately with β -arrestins at the early step of endocytosis, promoting ubiquitination, internalization, endosomal sorting and lysosomal degradation of activated GPCRs (Shea et al., 2012 [DOI](#)). The another study, however, proposed different hypothesis suggesting that α -arrestins might act as secondary adaptor localized at endosomes to mediate endosomal sorting of cargo molecules (S. O. Han et al., 2013 [DOI](#)). Among the protein complexes that interact with α -arrestins, we identified those related with clathrin coated pit in human and AP-2 adaptor complex in *Drosophila* (Figure 2 [DOI](#)). They are multimeric proteins to induce internalization of cargo molecules to mediate clathrin-mediated endocytosis, which suggests involvement of α -arrestins in early step of endocytosis.

Among the interacting proteins, 58 orthologous interacting groups were observed to be conserved between human and *Drosophila*, suggesting conserved roles of α -arrestins between two species (Figure 3 [DOI](#)). Among conserved proteins, proteins known to interact with human α -arrestins, such as NEDD4, WWP2, WWP1, and ITCH, were identified along with its orthologs in *Drosophila*, which are Su(dx), Nedd4, and Smurf, implying that regulatory pathway of ubiquitination-dependent proteolysis by α -arrestins is also present in invertebrate species. Besides the known conserved functions, the novel conserved functions of α -arrestins interactomes were also identified, such as RNA splicing (Figure 1E [DOI](#); Figure 3 [DOI](#)). Because our protocol did not include treatment with RNase before the AP/MS, it is possible that RNA binding proteins could co-precipitate with other proteins that directly bind to α -arrestins through RNAs, and thus could be indirect binding partners. Nevertheless, other RNA binding proteins except for RNA splicing and processing factors were not enriched in our interactomes, indicating that this possibility may be not the case. Thus, it might be of interest to explore how α -arrestins are linked to RNA processing in future.

Some protein complexes and functional modules were found to be involved in specific cellular processes discovered in only human, suggesting that some functional roles of α -arrestins have diverged through evolution. As examples of specific cellular functions of α -arrestins, we explored the biological relevance of two interacting protein complexes: 1) the interaction between TXNIP and chromatin remodelers and 2) the interaction between ARRDC5 and the V-type ATPase complex. Given that TXNIP interacts with chromatin remodelers, such as the HDAC and histone H2B ubiquitination complexes, we speculated that chromatin structures could be affected by the interactions. Although we showed that siTXNIP treatment directed a global decrease in chromatin accessibilities and gene expression by inhibiting the binding of HDAC2 to targets, histones themselves could be also controlled by the interaction between TXNIP and the H2B ubiquitination complex. An impact of TXNIP on histone ubiquitination could strengthen the negative regulation of target loci by siTXNIP treatment. In addition, TXNIP interacts with the proteasome, which induces the degradation of binding partners (Figure 2 [DOI](#)). However, we observed that the cellular level and localization of HDAC2 were not affected by TXNIP reduction (Figure 5A [DOI](#) and B [DOI](#); Figure S7A [DOI](#)), meaning that the proteasome seems not to be involved in TXNIP's influence on HDAC2; rather, TXNIP directly hinders HDAC2 recruitment to target loci.

Because the V-type ATPase plays a key role in osteoclast differentiation and physiology (Feng et al., 2009 [↗](#); Qin et al., 2012 [↗](#)), we investigated a possible role for the ARRDC5-V-type ATPase interaction in this cell type. We showed that the ectopic expression of ARRDC5 increased both the differentiation of osteoclasts into their mature form and their bone reabsorption activity. Additionally, ARRDC5 co-localized with the V-type ATPase at the PM (**Figure 6C** [↗](#)). Thus, further characterization of ARRDC5 and its interactome in osteoclasts might clarify how ARRDC5 regulates the V-type ATPase to play a role in osteoclast differentiation and function. With the results, the discovery of new binding partners and their functions of TXNIP and ARRDC5 will facilitate the further investigations to explore the novel PPIs of α -arrestins.

Given the plethora of PPIs uncovered in this study, we also anticipate that our study could provide insight into many disease models. In fact, despite a limited knowledge of their biology, α -arrestins have already been linked to a range of cellular processes and several major health disorders, such as diabetes (Batista et al., 2020 [↗](#); Wondafrash et al., 2020 [↗](#)), cardiovascular diseases (Domingues, Jolibois, Marquet de Rouge, & Nivet-Antoine, 2021 [↗](#)), neurological disorders (Tsubaki, Tooyama, & Walker, 2020 [↗](#)), and tumor progression (Y. Chen et al., 2020 [↗](#); Mohankumar et al., 2015 [↗](#); Oka et al., 2006 [↗](#)), making them potential therapeutic targets. For the community, we provide comprehensive α -arrestin interactome maps on our website (human: http://big.hanyang.ac.kr/alphaArrestin_Human [↗](#) and *Drosophila*: http://big.hanyang.ac.kr/alphaArrestin_Fly [↗](#)). Researchers can search and download their interactomes of interest as well as access information on potential cellular functions associated with these interactomes.

Materials and Methods

Generating *Drosophila* α -arrestin-GFP fusion DNA constructs

To create *Drosophila* ARRDC entry clones, we gathered cDNA sequences of twelve *Drosophila* α -arrestins : CG2993 (#2276, *Drosophila* Genomics Resource Center, DGRC, Bloomington, IN, USA), CG18744 (#1388606, DGRC), CG18745 (#12871, DGRC), CG18746 #9217, DGRC), CG18747 (#1635366, DGRC), CG18748 (#1387253, DGRC), CG2641 (#1649402, DGRC), CG10086 (#8816, DGRC), CG14696 (#1644977, DGRC), CG1105 (#4234, DGRC), Vdup1 (#1649326, DGRC), and Leash (Y. Kwon et al., 2013 [↗](#)). We then subcloned each cDNA sequence of *Drosophila* α -arrestins into pCR8 entry clone vector using pCR8/GW/TOPO TA cloning kit (#K250020, Thermo Fisher Scientific, Waltham MA, USA), by following manufacturer's protocol. To generate plasmids with suitable system for protein expression in *Drosophila* cell culture, we then subcloned these α -arrestins-containing-pCR8 plasmids into pMK33-Gateway-GFP destination vector (Y. Kwon et al., 2013 [↗](#); Kyriakakis, Tipping, Abed, & Veraksa, 2008 [↗](#)) using Gateway LR Clonase II enzyme mix (#11791020, Thermo Fisher Scientific), where coding sequences of α -arrestins are inserted before GFP sequence. Final constructs were validated by GENEWIZ Sanger Sequencing.

Establishing *Drosophila* α -arrestin-GFP stably expressing cell lines

S2R+ cells were maintained in Schneider's *Drosophila* Medium (#21720024, Thermo Fisher Scientific) supplemented with 10% heat inactivated FBS (#16140071, Thermo Fisher Scientific) and 1% Penicillin Streptomycin (#15070063, Thermo Fisher Scientific) at 24°C. To establish α -arrestin-GFP stably expressing *Drosophila* cell lines, 0.4×10^6 S2R+ cells were seeded in 6-well plates and were transfected with 1 μ g of each pMK33-ARRDC-GFP construct using Effectene transfection reagent (#301425, Qiagen, Venlo, Netherlands). pMK33 plasmid is a copper-induced protein expression vector, which carries Hygromycin B-antibiotic-resistant gene. Therefore, we selected for α -arrestin-GFP stable cell lines by maintaining cells in Schneider's *Drosophila* Medium supplemented with 200 μ M Hygromycin B (#40-005, Fisher Scientific). The stable cells were transferred into T25 cm² flasks to repopulate. To induce the expression of α -arrestin-GFP fusion

proteins, we exposed the stable cells to 500 μM CuSO_4 (#C8027, Sigma Aldrich, Burlington, MA, USA) to the media. We confirmed the GFP-tagged α -arrestin protein expressions using fluorescence microscopy.

Synthesizing human α -arrestin coding sequence

Due to the lack of commercially available stock, we utilized GENEWIZ (South Plainfield, NJ, USA) gene synthesis service to synthesize human ARRDC5 coding sequence (NM_001080523).

Generating mammalian GFP- α -arrestin fusion DNA constructs

To create human α -arrestin entry clones, we subcloned ARRDC3 (#38317, Addgene, Watertown, MA, USA) and ARRDC5 (GENEWIZ) into pCR8 entry clone vector using pCR8/GW/TOPO TA cloning kit (#K250020, (Thermo Fisher Scientific), by following manufacturer's protocol. ARRDC1 (BC032346, GeneBank), ARRDC2 (BC022516, GeneBank), ARRDC4 (BC070100, GeneBank), and TXNIP (BC093702, GeneBank) were cloned into pCR8. To generate plasmids with suitable system for protein expression in mammalian cell culture, we then subcloned these α -arrestin s-containing-pCR8 plasmids into pHAGE-GFP-Gateway destination vector (gift from Dr. Chanhee Kang at Seoul National University) using Gateway LR Clonase II enzyme mix (#11791020, Thermo Fisher Scientific), where coding sequences of α -arrestin are inserted after GFP sequence. Final constructs were validated by GENEWIZ Sanger Sequencing.

Establishing mammalian GFP- α -arrestin stably expressing cell lines

We produced GFP- α -arrestins lentiviral particles by seeding 5×10^6 HEK293T cells in 10 cm^2 dish with Dulbecco's Modified Eagle Medium (#11965118, Thermo Fisher Scientific) supplemented with 25 mM HEPES, 10% heat-inactivated fetal bovine serum (#16140071, Thermo Fisher Scientific), and 1% Penicillin Streptomycin (#15070063, Thermo Fisher Scientific) at 37°C in humidified air with 5% CO_2 . Approximately after 16-24 hours (hr), at 90% cell confluency, we changed the cell media to Opti-MEM medium (#31985070, Thermo Fisher Scientific) and transfected the cells with 10 μg pHAGE-GFP- α -arrestin construct, 10 μg lentivirus packaging plasmid (pCMV-dR8.91), and 10 μg virus envelope plasmid (VSV-G) using PEIPro DNA transfection reagent (#115010, VWR, Radnor, PA, USA). GFP- α -arrestins lentiviral particles were harvested 40 hr-post transfections. To establish GFP- α -arrestins stably expressing mammalian cell lines, HEK293 cells were seeded in 10 cm^2 dish with Dulbecco's Modified Eagle Medium (#11965118, Thermo Fisher Scientific) supplemented with 25 mM HEPES, 10% heat-inactivated fetal bovine serum (#16140071, Thermo Fisher Scientific) and 1% Penicillin Streptomycin (#15070063, Thermo Fisher Scientific) at 37°C in humidified air with 5% CO_2 . At 90% cell confluency, cells were infected with pHAGE-GFP- ARRDC lentivirus particle, and stable cells were selected by maintaining cells in media supplemented with 1.5 $\mu\text{g}/\text{mL}$ puromycin (#BP2956100, Thermo Fisher Scientific). We confirmed the GFP-tagged α -arrestin protein expressions using fluorescence microscopy.

Affinity purification of *Drosophila* and human GFP-tagged α -arrestin complexes

We seeded each of the *Drosophila* α -arrestin-GFP stable cells in six T-75 cm^2 flasks (2.1×10^6 cells per flask) and α -arrestin-GFP expression was induced for 48 hr with 500 μM CuSO_4 . Meanwhile, we seeded each of the human GFP- α -arrestin stable cells in eight T-75 cm^2 flasks and grown for 48 hr before collection. The cells were harvested by spinning down cells at 1,000g for 5 minutes (min) and washed once with cold PBS. We lysed the cells by resuspending cells in lysis buffer (10 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.5 mM EDTA, 1.5 mM MgCl_2 , 5% glycerol, 0.5% NP-40, 25 mM NaF, 1mM DTT, and 1x HALT protease and phosphatase inhibitor (#PI78442, Thermo Fisher Scientific)) and incubating them for 40 min. The lysate was separated from the insoluble fraction by centrifugation at 20,000 g for 15 min at 4°C. To capture the α -arrestins and their native interactors,

each α -arrestin-containing lysate was incubated with GFP-nanobody-conjugated to Dynabeads M-270 Epoxy magnetic beads (#14301, Thermo Fisher Scientific). The supernatant was separated from the beads using a magnetic rack, and the beads were washed five times with lysis buffer. The protein complexes were eluted from the beads by adding 200 mM glycine pH 2.5 and the pH was neutralized with Tris base pH 10.4. Purified α -arrestin proteins were confirmed by running a fraction of the eluted proteins on SDS-PAGE/Coomassie gel.

Protein sample preparation for mass spectrometry

To digest protein samples into peptides for mass spectrometry analysis, we precipitated the eluted proteins by adding trichloroacetic acid (#T0699, Sigma Aldrich) to 20% final concentration, followed by spinning down samples at maximum speed for 30 min at 4°C. The precipitates were washed with 10% trichloroacetic acid solution and three additional times with Acetone (#A929, Thermo Fisher Scientific), and left to dry in room temperature. Protein precipitations were digested with Trypsin (Promega, #V5113) diluted in Digestion buffer (100 mM Ammonium Bicarbonate and 10% Acetonitrile) in 1:40 ratio. Resulting peptides were purified using ZipTip Pipet tips (#ZTC18M096, Thermo Fisher Scientific).

LC/MS-MS analysis

We used cells stably expressing GFP and wild-type HEK293 or S2R+ cells alone as control baits. AP/MS experiments for all *Drosophila* and human α -arrestin baits were performed in two biological replicates, with the exception of human ARRDC3 baits (two technical replicates). Samples were resuspended in Mass Spectrometry buffer (5% Formic Acid and 5% Acetonitrile) and were analyzed on an Liquid Chromatography Orbitrap Fusion Lumos Tribrid Mass Spectrometer (#IQLAAEGAAPFADMBHQ, Thermo Fisher Scientific) equipped with a nano-Acquity UPLC system and an in-house developed nano spray ionization source. Peptides were separated using a linear gradient, from 5-30% solvent B (LC-MS grade 0.1% formic acid (#A117, Thermo Fisher Scientific) and acetonitrile) in a 130 min period at a flow rate of 300 nL/min. The column temperature was maintained at a constant 5°C during all experiments. Peptides were detected using a data dependent method. Survey scans of peptide precursors were performed in the Orbitrap mass analyzer from 380 to 1500 m/z at 120K resolution (at 200 m/z) with a 5 x 10⁵ ion count target and a maximum injection time of 50 milliseconds (ms). The instrument was set to run in top speed mode with 3 seconds (sec) cycles for the survey and the MS/MS scans.

Identification of high-confidence bait-prey PPIs

SAINTexpress analysis

To identify high-confidence bait-prey PPIs, spectral counts of AP/MS data from S2R+ and HEK293 cells were subjected to the SAINTexpress algorithm (version 3.6.1) (Teo et al., 2014 [DOI](#)), which calculates the probability of authenticity for each bait prey PPI. The program outputs the SAINTexpress scores and the Bayesian false discovery rates (BFDR) based on the spectral count distribution of true and false PPI sets. Before calculating the scores, bait-to-bait self-interactions were removed manually. SAINTexpress was run with the “-R 2” parameter, which specifies the number of replicates, and the “-L 3” parameter, which specifies the number of representative negative control experiments to be considered.

PPI validation datasets

To evaluate the performance of the PPI prediction based on the SAINTexpress score, validation datasets including positive and negative PPIs were precompiled as described in previous studies (Y. Kwon et al., 2013 [DOI](#); Vinayagam et al., 2016 [DOI](#)). Briefly, the positive PPIs were initially collected by searching for known PPIs involving α -arrestins from STRING version 10.5 (Szklarczyk et al., 2015 [DOI](#)), GeneMANIA version 3.4.1 (Warde-Farley et al., 2010 [DOI](#)), Bioplex (Huttlin et al., 2015 [DOI](#)),

and DpiM (Guruharsha et al., 2011). For human, additional positive PPIs were curated from the literature (Colland et al., 2004; Dotimas et al., 2016; Nabhan et al., 2012; Nishinaka et al., 2004; Puca & Brou, 2014; Wu et al., 2013). After these steps, 30 PPIs (21 preys) for human and 46 PPIs (17 preys) for *Drosophila* were considered as positive PPIs (Tables S3A and C). Proteins manually curated from the Contaminant Repository for Affinity Purification (CRAPome) (Mellacheruvu et al., 2013) were compared to those detected in our negative controls and only those that were detected in both were considered as negative PPIs (Tables S3B and D). As a result of these steps, 1,372 PPIs (268 preys) for human and 1,246 PPIs (122 preys) for *Drosophila* were compiled as negative PPIs.

Construction of high-confidence PPI networks

The performance of SAINTexpress was evaluated using the positive and negative PPIs. Because there is an imbalance between positive and negative PPIs, 1000 random cohorts of negative PPIs number-matched with that of positive PPIs were generated. The average true positive and false positive rates were plotted as ROC curves over different SAINTexpress scores as a cutoff, and AUC values were calculated using the ROC R package (version 1.0-11, <https://cran.r-project.org/web/packages/ROCR>). Based on these results, we chose an optimal cutoff for high confidence PPIs with a BFDR of 0.01, where the false positive rates were less than 3% (~1.8 % for human and ~2.7% for *Drosophila*) in both species, and the true positive rates were substantially higher (~66.7 % for human and ~45.7% for *Drosophila*). The cutoffs correspond to SAINTexpress scores of 0.85 and 0.88 for human and *Drosophila*, respectively.

Identification of protein complexes associated with α -arrestins

To examine protein complexes significantly enriched in the α -arrestin interactomes, we collected known protein complexes from two databases: COMPLEAT (Vinayagam et al., 2013), which is a comprehensive resource of protein complexes built from information in the literature and predicted by orthologous relationships of proteins across species (human, *Drosophila*, and yeast), and the DAVID GO analysis of cellular components (Huang et al., 2009a, 2009b) (Benjamini-Hochberg FDR ≤ 0.05) (Table S7B and E), from which bulk cellular compartments such as the nucleus, cytosol, and so on were excluded. From the COMPLEAT database, we evaluated the association of the resulting protein complexes with each α -arrestin by the complex association score, which is the IQM of SAINTexpress scores (Equation 1)

$$\text{Complex association score (IQM)} = \frac{\sum_{i=Q1}^{Q3} \text{SAINTexpress score}_i}{(Q3-Q1)+1} \quad [\text{Equation 1}]$$

, where the first quartile is $Q1 = \frac{N}{4} + 1$ the third quartile is $Q3 = \frac{3N}{4}$, and N is the total number of preys in the complex. The significance of the complex association score was estimated by comparing the score to the null distribution of the scores calculated from 1,000 random complexes of input proteins. The significance was tested through the online COMPLEAT tool, and protein complexes with $P < 0.05$ were selected for further analysis (Table S7A and D). Next, we iteratively combined (clustered) the pairs of protein complexes from any two databases (COMPLEAT and GO analysis of cellular components) that showed the highest overlap coefficients, $\text{Overlap}(X,Y)$ (Equation 2) (Vijaymeena & Kavitha, 2016), until there was no pair of complexes whose coefficients were higher than 0.5.

$$\text{Overlap}(X,Y) = \frac{|X \cap Y|}{(|X|, |Y|)} \quad [\text{Equation 2}]$$

From the clustered set of complexes, we manually removed those with fewer than three subunits or two PPIs. Subunits in the complexes that have no connection among themselves were also removed. Lastly, the significance of associations of the resulting complexes with each α -arrestin

were tested in the same manner as done in COMPLEAT using complex association score. The resulting P values were corrected by the Benjamini-Hochberg procedure and only interactions with statistical significance (FDR < 0.05) were visualized with Cytoscape v3.5.1 (Shannon et al., 2003) (Figure 2).

Orthologous networks of α -arrestin interactomes

DIOPT (version 7.1) was used to search for orthologs of all prey proteins and only those with a DIOPT score ≥ 2 were selected for the identification of orthologous PPIs between *Drosophila* and human. Next, the orthologs were tested for the enrichment of GO biological process and molecular functions and Kyoto Encyclopedia of Genes and Genomes pathway using the DAVID (Huang et al., 2009a, 2009b). In addition, manual curation of individual genes was performed through the Uniprot database (UniProt Consortium, 2018). The orthologs were manually grouped into functional modules based on the results and α -arrestins were modularized into seven groups based on hierarchical clustering of log2-transformed mean spectral counts using the correlation distance and the Ward linkage method. The heatmap was plotted using the pheatmap R package (version 1.0.12).

TXNIP knockdown in HeLa cells

HeLa cells (CCL-2; ATCC, Manassas, VA, USA) were cultured in complete DMEM supplemented with 10% FBS and 1% penicillin-streptomycin. Cells were cultured in an incubator at 37°C in humidified air containing 5% CO₂. For siRNA-induced knockdown of TXNIP in HeLa cells, the following siRNA duplex was synthesized (Bioneer, Daejeon, South Korea): sense: 5'-GUCAGUCACUCUCAGCCAuTdT-3', anti-sense: 5'-AUGGCUGAGAGU GACUGACdTdT-3'. Random sequence siRNAs (AccuTarget Negative control siRNA; Bioneer), which are non-targeting siRNAs that have low sequence homology with all humans, mouse, and rat genes, were used as negative controls (siCon). 100 nM of each siRNA was transfected into 105 HeLa cells using Lipofectamine RNAiMAX (#13778075, Invitrogen, Carlsbad, CA, USA; Thermo Fisher Scientific) according to the manufacturer's instructions. Transfected cells were harvested after 48 hr for RNA-seq and ATAC-seq (two biological replicates for each sequencing data).

RNA sequencing

For RNA-seq, total RNA was extracted using TRIzol (#15596018, Invitrogen; Thermo Fisher Scientific) according to the manufacturer's protocol. Total RNA concentration was calculated by Quant-IT RiboGreen (#R11490, Invitrogen; Thermo Fisher Scientific). To assess the integrity of the total RNA, samples are run on the TapeStation RNA screentape (#5067- 5576, Agilent Technologies, Santa Clara, CA, USA). Only high-quality RNA preparations, with RNA integrity number greater than 7.0, were used for RNA library construction. A library was independently prepared with 1 μ g of total RNA for each sample by Illumina TruSeq Stranded mRNA Sample Prep Kit (#RS-122-2101, Illumina, Inc., San Diego, CA, USA). The first step in the workflow involves purifying the poly-A containing mRNA molecules using poly-T-attached magnetic beads. Following purification, the mRNA is fragmented into small pieces using divalent cations under elevated temperature. The cleaved RNA fragments are copied into first strand cDNA using SuperScript II reverse transcriptase (#18064014, Invitrogen, Thermo Fisher Scientific) and random primers. This is followed by second strand cDNA synthesis using DNA Polymerase I, RNase H and dUTP. These cDNA fragments then go through an end repair process, the addition of a single 'A' base, and then ligation of the adapters. The products are then purified and enriched with PCR to create the final cDNA library. The libraries were quantified using KAPA Library Quantification kits (#KK4854, KAPA BIOSYSTEMS, Wilmington, MA, USA) for Illumina Sequencing platforms according to the qPCR Quantification Protocol Guide and qualified using the TapeStation D1000 ScreenTape (#5067- 5582, Agilent Technologies). Indexed libraries were then submitted to an Illumina NovaSeq 6000 (Illumina, Inc.) as the paired-end (2 $\times$ 100 bp) sequencing. Both library preparation and sequencing were performed by the Macrogen (Macrogen, Inc., Seoul, South Korea).

ATAC sequencing

100,000 cells were prepared using LUNA-FL™ Automated Fluorescence Cell Counter (#L20001, logox biosystems, Gyeonggi-do, South Korea). Cells were lysed using cold lysis buffer, which consist of Nuclease-free water (#10977023, Invitrogen; Thermo Fisher Scientific), IGEPAL CA-630 (#18896, Sigma Aldrich), 1M Trizma HCl(PH7.4) (#T2194, Sigma Aldrich), 5M NaCl (#59222C, Sigma Aldrich), and 1M MgCl₂ (#M1028, Sigma Aldrich). The nuclei concentration was determined using Countess II Automated Cell Counter (#AMQAX1000, Thermo Fisher Scientific) and nuclei morphology was examined using microscopy. Immediately after lysis, resuspend nuclei (50,000 cells) were put in transposition reaction mix 50 µl, which consist of TED1 2.5µl and TD 17.5 µl (#20034197, Illumina, Inc.), nuclease free water 15 µl, and the nuclei resuspension (50,000 nuclei, 15 µl). The transposition reaction was incubated for 30 min at 37°C. Immediately following transposition, the products were purified using a MinElute PCR purification Kit (#28004, Qiagen). Next, transposed DNA fragments were amplified using Nextera DNA Flex kit (#20018704, Illumina, Inc.). To reduce GC and size bias in PCR, the appropriate number of cycles was determined as follows: qPCR side reaction was run, the additional number of cycles needed were calculated, liner Rn versus cycle was plotted and the cycle number that corresponds to 1/4 of maximum fluorescent intensity was determined. The remaining PCR reaction was run to the cycle number determined. Amplified library was purified and then quantified using KAPA library quantification kit (#07960255001, Roche, Basel, Switzerland) and Bioanalyzer (Agilent technologies). The resulting libraries were sequenced using HiSeq X Ten (Illumina, Inc.). Both library preparation and sequencing were performed by the Macrogen (Macrogen, Inc).

Processing of RNA-seq data

For quality checks and read trimming, RNA-seq data were processed by FastQC (version 0.11.8) (Andrews, 2010 [↗](#)) and sickle (version 1.33) (Joshi NA, 2011 [↗](#)) with default parameters. After the trimming, the reads were aligned to human transcriptomes (GENCODE version 29, GRCH38/hg38) (Frankish et al., 2019 [↗](#)) using STAR (version 2.5.3a_modified) (Dobin et al., 2013 [↗](#)) with default parameters and read counts were determined using RSEM (version 1.3.1) (Li & Dewey, 2011 [↗](#)). The DEG analysis was performed using the edgeR R package (version 3.32.1) (Robinson, McCarthy, & Smyth, 2010 [↗](#)). Batch information was added as confounding variables to adjust for batch effects. The DEGs are summarized in Supplementary Table S10.

Processing of ATAC-seq data

Each ATAC-seq dataset was processed using the ENCODE ATAC-seq pipeline implemented with Caper (<https://github.com/ENCODE-DCC/atac-seq-pipeline> [↗](#)) (Jin Lee, 2016). Briefly, reads were mapped to the human reference genome (GRCH38/hg38) using Bowtie2 (version 2.3.4.3), and unmapped reads, duplicates, and those mapped to the mitochondrial genome were removed. Peaks were called by MACS2 (Zhang et al., 2008 [↗](#)) and optimal peaks that were reproducible across pseudo replicates were used in the downstream analysis. The numbers of processed reads and peaks are summarized in Table S9. Plots of ATAC-seq signals around the TSSs of expressed genes were generated by the R genomation package (version 1.22.0) (Akalin, Franke, Vlahovicek, Mason, & Schubeler, 2015 [↗](#)). The batch effects of the signals were corrected by the removeBatchEffect function from the limma R package (version 3.46.0) (Ritchie et al., 2015 [↗](#)). Of the broad and narrow peaks resulting from the ENCODE ATAC-seq pipeline, the latter were used as an input to obtain consensus ACRs using the diffBind R package (version 3.0.15) (Ross-Innes et al., 2012 [↗](#)). The dACRs were detected using the edgeR R package (version 3.32.1) (Robinson et al., 2010 [↗](#)). In total, 70,746 ACRs and 5,219 dACRs were detected in HeLa cells and are summarized in Supplementary Table S10. The genomic positions of the ACRs were annotated through the ChIPseeker R package (version 1.26.2) (Yu, Wang, & He, 2015 [↗](#)). If the ACRs spanned more than one genomic region, their positions were assigned based on the following priority: promoters > 5' untranslated regions (UTRs) > 3'UTRs > other exons > introns > downstream > intergenic regions. The promoter of a gene was defined as the region 5 kb upstream and 500 bp downstream of the TSS.

Nuclear-cytoplasmic fractionation

Prior to transfection, HeLa cells were seeded in 100 mm cell culture dishes containing Opti-MEM medium and incubated overnight (reaching a confluency of approximately 30%- 40%). The cells were then transfected with siTXNIP. Cells were harvested after 48 hr of transfection and fractionated according to the manufacturer's instructions using NE-PER Nuclear and Cytoplasmic Extraction Reagents (#78833, Thermo Fisher Scientific). Protease inhibitor cocktail (P8340; Sigma Aldrich) was added as a supplement to the lysis buffer and the protein concentration was measured using a Pierce BCA Protein Assay Kit (#23225, Thermo Fisher Scientific).

Chromatin Immunoprecipitation (ChIP) Assay

Cells were crosslinked with 1% formaldehyde at 37°C or room temperature for 15 min and the reaction was stopped by the addition of 0.125M glycine. ChIP was then performed using a ChIP-IT High Sensitivity kit (#53040, Active Motif, Carlsbad, CA, USA) according to the manufacturer's instructions. Enrichment of the ChIP signal was detected by quantitative real-time PCR (qPCR). The data of each biological replicate were normalized with negative control IgG signals and enrichment values were calculated using the $\Delta\Delta C_t$ method (Helleman, Mortier, De Paepe, Speleman, & Vandesompele, 2007 [\[1\]](#)). The following antibodies were used: TXNIP (14715, Cell Signaling Technology, Beverly, MA), HDAC2 (57156, Cell Signaling Technology), H3ac antibody (39139; Active Motif, Carlsbad, CA), and normal rabbit IgG antibodies were used. The primers used for ChIP-qPCR are summarized in Table S12.

Osteoclast differentiation and collection of lentiviruses for ARRDC5 expression

BMMs were cultured as previously described (S. Y. Kim et al., 2019 [\[2\]](#)). Briefly, bone marrow was obtained from mouse femurs and tibias at 8 weeks of age, and BMMs were isolated from the bone marrow using Histopaque (1077; Sigma Aldrich). BMMs were seeded at a density of 1.2×10^5 cells/well into 24-well culture plates and incubated in α -MEM (SH30265.01; Hyclone, Rockford, IL, USA) containing 20 ng/mL macrophage colony stimulating factor (M-CSF) (300-25; PeproTech, Cranbury, NJ, USA). To induce osteoclast differentiation, BMMs were treated for 24 hr with lentiviral-containing medium that also contained M-CSF, after which the medium was changed to α -MEM containing 20 ng/mL M-CSF and 20 ng/mL RANKL (462-TEC; R&D Systems, Minneapolis, MN, USA). The differentiation medium was changed every 24 hr during the 5-day differentiation period.

To obtain the media containing lentivirus, HEK293 cells were cultured in DMEM containing 4.5 g/L glucose (SH30243.01; Hyclone) supplemented with 10% FBS (SH30084.03; Hyclone) and 1% penicillin-streptomycin. After seeding cells at a density of 1×10^5 cells/well into 6-well culture plates, the cells were incubated with lentivirus co-transfected media for 16 hr. Lentivirus co-transfected media was prepared according to the manufacturer's instructions using the CRISPR & MISSION® Lentiviral Packaging Mix (SHP002; Sigma Aldrich) and the lentiviral transfer vector, pHAGE-GFP-GFP or pHAGE-GFP-ARRDC5. After the incubation, the medium was replaced with fresh α -MEM medium supplemented with 10% FBS and 1% penicillin-streptomycin. The medium was collected twice (after 24 and 48 hr), designated as lentiviral-containing medium, and stored in a deep freezer until used to infect BMMs.

TRAP staining and bone resorption pit assay

Osteoclast differentiation and activity were determined by TRAP staining and a bone resorption pit assay, respectively. TRAP staining was performed using a TRAP staining kit (PMC-AK04F-COS; Cosmo Bio Co., LTD., Tokyo, Japan) following the manufacturer's instructions. TRAP-positive multinucleated cells with more than three nuclei were counted under a microscope using ImageJ software (NIH, Bethesda, MD, USA). The bone resorption pit assay was performed using dentin

discs (IDS AE-8050; Immunodiagnostic Systems, Tyne & Wear, UK). Cells were differentiated to osteoclasts on the discs over a 4-day period, after which the discs were stained with 1% toluidine blue solution and the resorption pit area was quantified using ImageJ software.

Immunofluorescence staining of the V-type ATPase and visualization with GFP ARRDC5

To inhibit V-type ATPase transport to the membrane (Matsumoto et al., 2019), osteoclasts on the fifth day of differentiation were incubated with 100 nM bafilomycin A1 (19-148; Sigma Aldrich) for 3 hr. Then, immunofluorescence staining was performed to visualize the localization of the V-type ATPase in bafilomycin A1-treated and untreated cells. The cells were fixed using a 4% paraformaldehyde solution (PC2031-100; Biosesang, Gyeonggi-do, Korea) and permeabilized using 0.05% Triton X-100 at room temperature for 5 min. The cells were incubated with anti-V-type ATPase antibody (SAB1402125-100UG; Sigma Aldrich) at room temperature for 1 hr, and then stained with the Alexa Fluor 594-conjugated anti-mouse antibody (A-21044; Invitrogen) at room temperature for 30 min. Finally, cells were mounted using Antifade Mountant with DAPI (P36962; Invitrogen). Fluorescence images were observed under a ZEISS confocal microscope (LSM5; Carl Zeiss, Jena, Germany).

Data availability

All AP/MS raw tables from human and *Drosophila* α -arrestins are deposited in Source Data. ATAC-seq and RNA-seq data from HeLa cells treated with siCon and siTXNIP, can be downloaded from the Korean Nucleotide Archive (KONA; PRJKA220517, <https://www.kobic.re.kr/kona/>).

Code availability

All source codes and in-house codes used in the study are available at GitHub (https://github.com/Kyung-TaeLee/alphaArrestins_PPI_network).

Acknowledgements

We thank all of the BIGLab and Kwon Lab members for critical reading and comments. This work was supported by the National Research Foundation (NRF) funded by the Ministry of Science & ICT (2020R1A4A1018398, 2021R1A2C3005835, and 2022M3E5F1018502 to JWN, and 2020R1A6A3A13077354 to KTL).

Author contributions

KTL performed all computational analyses; IP performed all AP/MS experiments; SYK contributed to ARRDC5-related functional study, HJC and NBT performed ChIP-related experiments, and NBT and HSC performed ATAC-seq and RNA-seq experiments; KTL contributed to writing the codes; KTL, IP, NBT, HJC, JEK, YK, and JWN contributed to writing the manuscript; YK and JWN supervised the project and conceived the idea.

Supplementary Methods

Immunofluorescence imaging of human α -arrestins

Stably α -arrestin-GFP expressing HEK293 cells were cultured in a 12 well-plate with pre-sterilized round glass coverslips in each well. Cells on coverslip were fixed in 4% paraformaldehyde (PFA) (RT15710, Electron Microscopy Sciences, Hatfield, PA, USA) diluted in PBS for 30 min and then washed three times with PBST (PBS supplemented with 0.2% Triton X-100) with 5 min intervals. To label the nucleus, samples were stained with DAPI (1:5000; D9542, Sigma Aldrich) in PBST supplemented with 1% BSA (A7906, Sigma Aldrich) for 1 hr at room temperature. Stained cells samples were washed three times with PBST and preserved in Vectashield (H-1000, Vector Laboratories, Burlingame, CA, USA). Fluorescence images were acquired using an Olympus FV1200 confocal microscope with 40X oil objective lens and 2X zoom factor. NIH ImageJ software was used for further adjustment and assembly of the acquired images.

Database searching and analysis of mass spectrometry data

MS/MS spectra were queried using the Comet search engine (*Eng et al., 2013* [link](#)) to search for corresponding proteins in Flybase (*Gramates et al., 2017* [link](#)) and Uniprot (*The UniProt, 2017* [link](#)). Common contaminant protein sequences from the Common Repository of Adventitious Proteins (cRAP) Database (<ftp://ftp.thegpm.org/fasta/cRAP> [link](#)) were used to filter contaminating sequences. Searching was done with following parameters: tryptic digest, internal decoy peptides, the number of missed cleavages=2, precursor tolerance allowing for isotope offsets=20 ppm, a 1.00 fragment bin tolerance, static modification of 57.02 on cysteine, and variable modification of 16.00 on methionine. The acetylation, phosphorylation, and ubiquitination searches add variable modifications of 42.01 on lysine, 79.97 on serine/threonine/tyrosine, and 114.04 on lysine, respectively. The search results were then processed through the Trans-Proteomic Pipeline suite of tools version 4.8.0 (*Keller et al., 2005* [link](#)) where the PeptideProphet tool (*Keller et al., 2002* [link](#)) was applied to calculate the probability that each search result is correct and the ProteinProphet tool (*Nesvizhskii et al., 2003* [link](#)) was applied to infer protein identifications and their probabilities.

Functional annotations and multiple sequence alignment of α -arrestin sequences

The sequences of twelve *Drosophila* and six human α -arrestins were retrieved from the Uniprot database (*UniProt Consortium, 2018* [link](#)). Domains and motifs including the PPxY motif were annotated based on sequences from Pfam version 31.0 (*El-Gebali et al., 2019* [link](#)) and the eukaryotic linear motif (ELM) database (*Dinkel et al., 2015* [link](#)). The sequences were subjected to the multiple-sequence alignment tool T-COFFEE (*Notredame et al., 2000* [link](#)) using default parameters. The output of T-COFFEE was applied to RAxML (version 8.2.11) (*Stamatakis, 2014* [link](#)) to generate a consensus phylogenetic tree with 1,000 rapid bootstrapping using “-m PROTGAMMAWAGF” as the parameter (<https://cme.h-its.org/exelixis/resource/download/> [link](#) NewManual.pdf).

Checking the reproducibility of spectral counts among replicates

If multiple proteins isoforms were detected, they were collapsed into a single gene. To avoid the divide-by-zero error, spectral counts of “0” were converted to a minimum non-zero value, “0.01”. To examine the integrity and quality of spectral counts from the AP/MS, the average correlation coefficients (Pearson) of spectral counts from α -arrestins were calculated and plotted. At each cutoff of spectra counts from 1 to 15, only the PPIs with spectral counts that were the same or higher than the cutoff for all replicates were kept and used to calculate correlation coefficients between replicates. The resulting coefficients from the α -arrestin interactomes were then

averaged and plotted. At the cutoff of 6 spectral counts, saturation of average correlation coefficients was observed and chosen as an optimal cutoff to filter the PPIs. Principal component analysis (PCA) of the filtered PPIs was conducted based on spectral counts (with a pseudo count 1 added) transformed into a $\log_2$ using the factoextra R package (version 1.0.7).

Hierarchical clustering of high-confidence PPIs

Hierarchical clustering based on $\log_2$ spectral counts (pseudo count 1 added) of high confidence PPIs was conducted using the Pearson correlation as the clustering distance and Ward's method as the clustering method. Heatmaps were visualized through the ComplexHeatmap R package (version 2.6.2) (Gu *et al.*, 2016 [DOI](#)). Six clusters were identified for each species based on the results of hierarchical clustering; the PANTHER protein class overrepresentation test was performed for the proteins in each cluster (Thomas *et al.*, 2003 [DOI](#)). False discovery rates (FDRs, Fisher's exact test) of indicated protein classes were ≤ 0.05 for all classes except for "GTPase-activating protein" in human (FDR < 0.133) and "GEFs" in *Drosophila* (FDR < 0.109), respectively. Interacting prey proteins from the positive PPIs were selectively labeled.

Domain and motif analysis of bait and prey proteins

For human and *Drosophila*, respectively, 53 and 65 short linear motifs in α -arrestins were annotated using the ELM database (Dinkel *et al.*, 2015 [DOI](#)), and 423 and 546 protein domains in prey proteins were annotated using the Uniprot database (UniProt Consortium, 2018 [DOI](#)) (Table S4). To test for enrichment of protein domains, we implemented the Expression Analysis Systematic Explorer (EASE) score (Hosack *et al.*, 2003 [DOI](#)), which is calculated by subtracting one gene within the query domain and conducting a one-sided Fisher's exact test. Protein domains enriched in the interactomes of each α -arrestin (Benjamini-Hochberg FDR ≤ 0.05) were plotted using the ComplexHeatmap R package (version 2.6.2). Next, to see how reliable our filtered PPIs were, we utilized information about known affinities between domains and short linear motifs from the ELM database (Dinkel *et al.*, 2015 [DOI](#)). Because the arrestin_N (Pfam ID : PF00339) and arrestin_C (Pfam Id : PF02752) domains in α -arrestins do not have known interactions with any of the short linear motifs in the ELM database (Dinkel *et al.*, 2015 [DOI](#)), only the interactions between the short linear motifs in α -arrestins and protein domains in the interactome (prey proteins) were considered in this analysis. We found that 59 out of the 390 human PPIs and 64 out of the 740 *Drosophila* PPIs were supported by such known affinities (Table S4). One-sided Fisher's exact test was used to test the significance of the enrichment of the supported PPIs in the filtered PPI sets versus those in the unfiltered PPI sets (Figure 1D [DOI](#)).

Subcellular localizations of bait and prey proteins

To search for annotated subcellular localizations of the proteins in the α -arrestin interactomes, we first obtained annotation files of cellular components (Gene Ontology (GO) : CC) for human and *Drosophila* from the Gene Ontology Consortium (Ashburner *et al.*, 2000 [DOI](#)). From the annotations, we only utilized GO terms for 11 subcellular localizations (name of subcellular localization – GO term ID: Cytosol – GO:0005829; Plasma membrane – GO:0005886; Nucleus – GO:0005634; Mitochondrion – GO:0005739; Endoplasmic reticulum – GO:0005783; Golgi apparatus – GO:0005794; Cytoskeleton – GO:0005856; Peroxisome – GO:0005777; Lysosome – GO:0005764; Endosome – GO:0005768; Extracellular space – GO:0005615). If a protein was annotated to be localized in multiple locations, a weighted value (1/the number of multiple localizations) was assigned to each location. Finally, the relative frequencies of the subcellular localizations associated with the interacting proteins in the filtered PPIs were plotted for each α -arrestin (Figure S3B [DOI](#)).

Immunoblotting and co-immunoprecipitation Assays

Cells were lysed in radioimmunoprecipitation assay (RIPA) buffer supplemented with protease inhibitor. For immunoblotting, the cell lysates were separated by 10% SDS polyacrylamide gel electrophoresis (PAGE) and transferred to nitrocellulose membranes. After blocking membranes with 5% skim milk in Tris buffered Saline containing 0.1% Tween-20 (TBS T) for 2 hours (hr) at room temperature, the nitrocellulose membranes were incubated with appropriate primary antibodies overnight at 4°C and subsequently reacted with horseradish peroxidase (HRP)-conjugated secondary antibodies for 1 hr at room temperature. Bands were visualized using an enhanced chemiluminescence (ECL) detection system, West-Q Pico ECL Solution (W3652-02, GenDEPOT, Katy, TX, USA). For quantification of immunoblot results, the densities of target protein bands were analyzed with Image J.

For immunoprecipitation, the cell lysates (2 mg) were incubated with appropriate antibodies (1 µg) overnight at 4°C and precipitated with TrueBlot Anti-Rabbit Ig IP agarose beads (Rockland, Philadelphia, PA) for 2 hr at 4°C. The immunocomplexes were washed with chilled PBS three times and heated with 3x sample loading buffer containing β-mercaptoethanol. The samples were separated by 6-8 % SDS-polyacrylamide gel electrophoresis (PAGE) and immunoblot was performed as described above.

The following antibodies were used for immunoblotting and co-immunoprecipitation assays: anti-TXNIP (#14715), anti-HDAC2 (#57156) and anti-alpha Tubulin (#3873) were obtained from Cell Signaling Technology (Beverly, MA); anti-H3ac (39139) was obtained from Active Motif (Carlsbad, CA); anti-β-actin (GTX629630) was obtained from GeneTex; normal anti-rabbit IgG (sc-2027) was obtained from Santa Cruz Biotechnology (Dallas, TX); TrueBlot anti-rabbit IgG HRP (18-8816-31) was obtained from Rockland (Philadelphia, PA).

Quantitative Reverse-transcription polymerase chain reaction (PCR)

Total RNA was isolated using TRIzol reagent (#15596018, Invitrogen, Carlsbad, CA, USA; Thermo Fisher Scientific) and subjected to reverse transcription PCR (RT-PCR) with ReverTra Ace qPCR RT kit (#FSQ-101, Toyobo, Osaka, Japan) or GoScript RT-PCR system (#A5001, Promega, Madison, WI, USA) according to the manufacturer's instructions. The mRNA expression levels of target genes were quantified using the CFX Opus 96 (Biorad, Hercules, CA) or Applied Biosystems QuantStudio 1 (Applied Biosystems, Foster city, CA) real time PCR. AccuPower 2X GreenStar™ qPCR Master Mix (#K6251, Bioneer, Daejeon, Republic of Korea) or SYBR Green Realtime PCR Master Mix (#QPK-201, Toyobo, Osaka, Japan) were applied according to the manufacturer's protocols. The data normalized by GAPDH or alpha tubulin mRNA levels and calculated using the $\Delta\Delta C_t$ method (Hellemans et al., 2007 [\[1\]](#)). The primers used for qRT-PCR analysis are summarized in Table S12.

PCA of ATAC- and RNA-seq data

For ATAC-seq, normalized read counts derived from the diffBind R package (version 3.0.15) (Ross-Innes et al., 2012 [\[2\]](#)) were transformed into a $\log_2$ function. Batch effect corrections were done using the limma R package (version 3.46.0) (Ritchie et al., 2015 [\[3\]](#)). For RNA-seq, counts per million mapped reads (CPM) were also processed in the same manner. For PCA, 2,000 features with the highest variance across samples were extracted and utilized. Plots of principal components 1 and 2 were generated by the factoextra R package (version 1.0.7).

Functional signatures of repressed genes upon TXNIP depletion

Genes that exhibited decreased chromatin accessibility at their promoter and decreased RNA expression upon TXNIP knockdown (Table S11) were selected based on the following criteria: 1. $\log_2$ (RNA level in siTXNIP-treated cells/RNA level in siCon-treated cells) (hereafter, siTXNIP/siCon)

≤ -1 ; 2. $\log_2$ (siTXNIP/siCon) of ACRs in the promoter region ≤ -1 (If there are multiple ACRs in the promoter region, the one with the highest ATAC-seq signal was selected) or $\log_2$ mean (siTXNIP/siCon) of all ACRs in the promoter region ≤ -1 . Enrichment analysis of the GO terms in the gene set was performed by g:Profiler (Raudvere et al., 2019). Top 10 enriched terms from the biological process and molecular functions categories were plotted (Figure 4G).

Immunofluorescence of HDAC2 and TXNIP

HeLa cells were cultured in 6-well plates with cover slips in each well (1.5×10^4 cells/well). After cells were incubated overnight in Opti-MEM, TXNIP knockdown was induced by transfection of siRNA at a concentration of 100 nM. Following 48 hr of transfection, the cells were washed twice with PBS and then fixed with 100% ice-cold methanol for 10 min at -20°C . After rinsing three times with PBSTw (PBS containing 0.1% Tween 20), the cells were blocked with 3% BSA in PBS and incubated for 45 min at room temperature. Next, cells were incubated with the primary antibody for 150 min followed by the secondary antibody for 60 min in the dark. For co staining with a second primary antibody, the blocking step followed by the primary and secondary antibody incubation steps were repeated. All of the antibodies were diluted in antibody dilution buffer (1% BSA in PBS). Information of the antibodies are listed in “antibody” section in STAR Method. The cover slips were rinsed three times with PBSTw and then mounted with VECTASHIELD Antifade Mounting Medium containing DAPI (Vector Laboratories, Newark, CA, USA) according to the manufacturer’s instructions. The fluorescence was visualized with a Nikon C2 Si-plus confocal microscope.

Supplementary figures

Supplementary Table and legends

Table S1. Information about α -arrestin proteins from human (A) and *Drosophila* (B).

Table S2. Positive and negative PPIs of α -arrestins for human (A and B) and *Drosophila* (C and D).

Table S3. Output tables of SAINTexpress run on AP/MS data for human (A) and *Drosophila* (B).

Table S4. Protein domains and short linear motifs annotated in the interactome of each α -arrestin of human (A and B) and *Drosophila* (D and E). Annotated interactions between the short linear motifs in α -arrestins and protein domains in the interactome from the ELM database are also summarized for human (C) and *Drosophila* (F).

Table S5. Results of enrichment test of Pfam domains in the interactome of each α -arrestin for human (A) and *Drosophila* (B).

Table S6. Summary tables of subcellular localizations assigned to proteins in the interactomes of α -arrestins for human (A) and *Drosophila* (B).

Table S7. Results of the protein complex enrichment analysis tool (COMPLEAT) and enrichment test of cellular component GO terms for human (A and B) and *Drosophila* (D and E). Protein complexes of two databases were iteratively clustered and are summarized in (C) and (F) for human and *Drosophila*, respectively. Information of clustered protein complexes from (C and F) were utilized to plot network of protein complexes in Figure 2.

Table S8. DIOPT predictions of orthologs of α -arrestins and proteins in their interactomes (A and B). Results of enrichment test of GO terms (biological process, molecular functions and BP, MF, and Kyoto Encyclopedia of Genes and Genomes pathway) are also summarized for orthologs in human (C) and *Drosophila* (D).

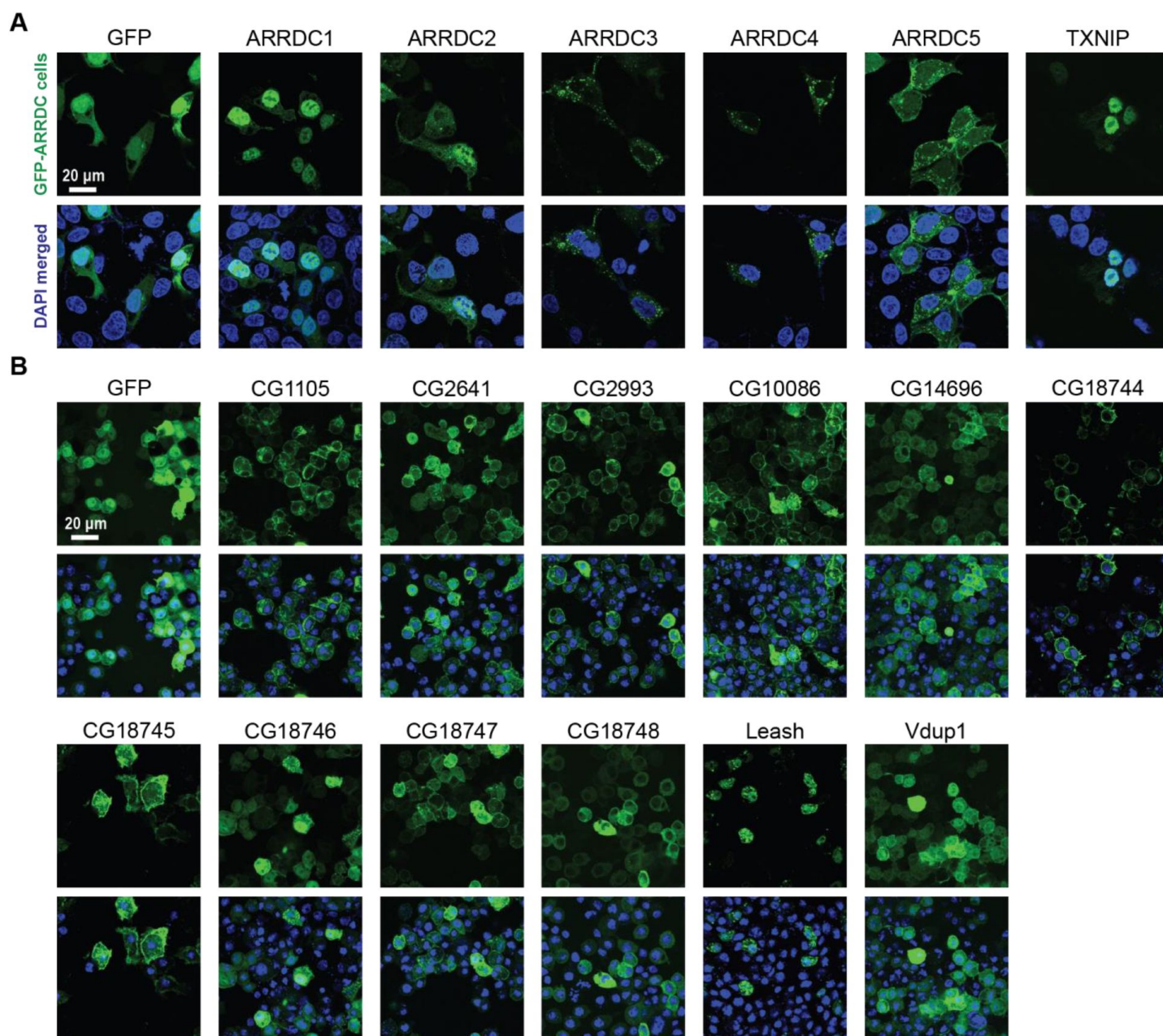


Figure S1.

Fluorescence images showing HEK293 and S2R+ cells stably expressing GFP tagged α -arrestins

Representative images of HEK293 (**A**) and S2R+ (**B**) cells stably expressing GFP-tagged α -arrestins.

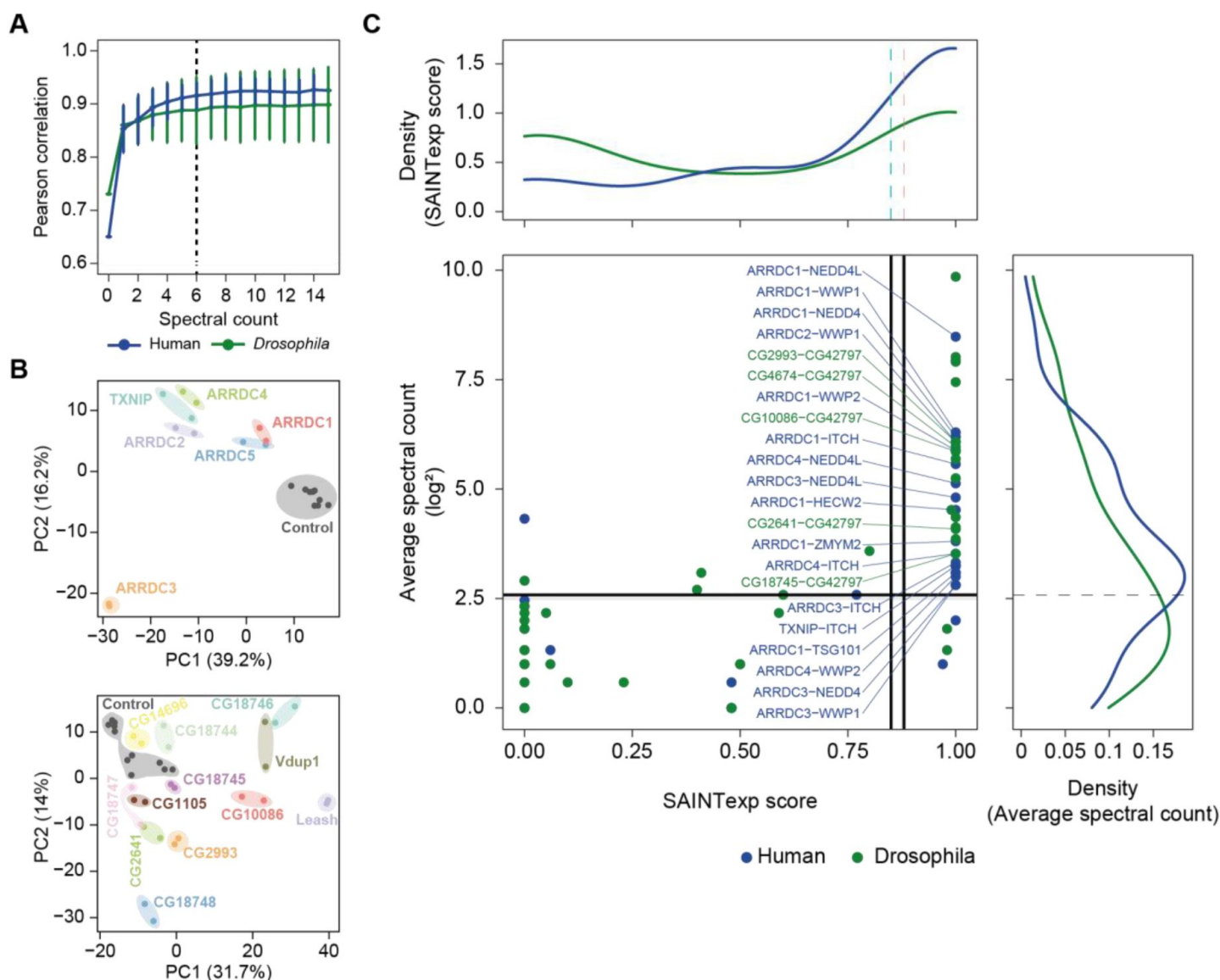


Figure S2.

Affinity purification / mass spectrometry (AP/MS) data are highly reproducible and recapitulate known PPIs

(A) Average Pearson correlation coefficients of $\log_2$ spectral counts between replicates of AP/MS of each α -arrestin at varying cutoffs are shown (mean $\pm$ standard deviation(sd)). The cutoff used in this study, 6, is shown as a dashed line. **(B)** PCA plots based on $\log_2$ spectral counts of high-confidence PPIs for human (upper) and *Drosophila* (lower) are shown. **(C)** SAINTExpress scores and average spectral counts ($\log_2$) of the positive PPIs (Table S2A and C) are shown and density plots for each axis are also plotted. The positive PPIs that are included in the filtered set are selectively labeled.

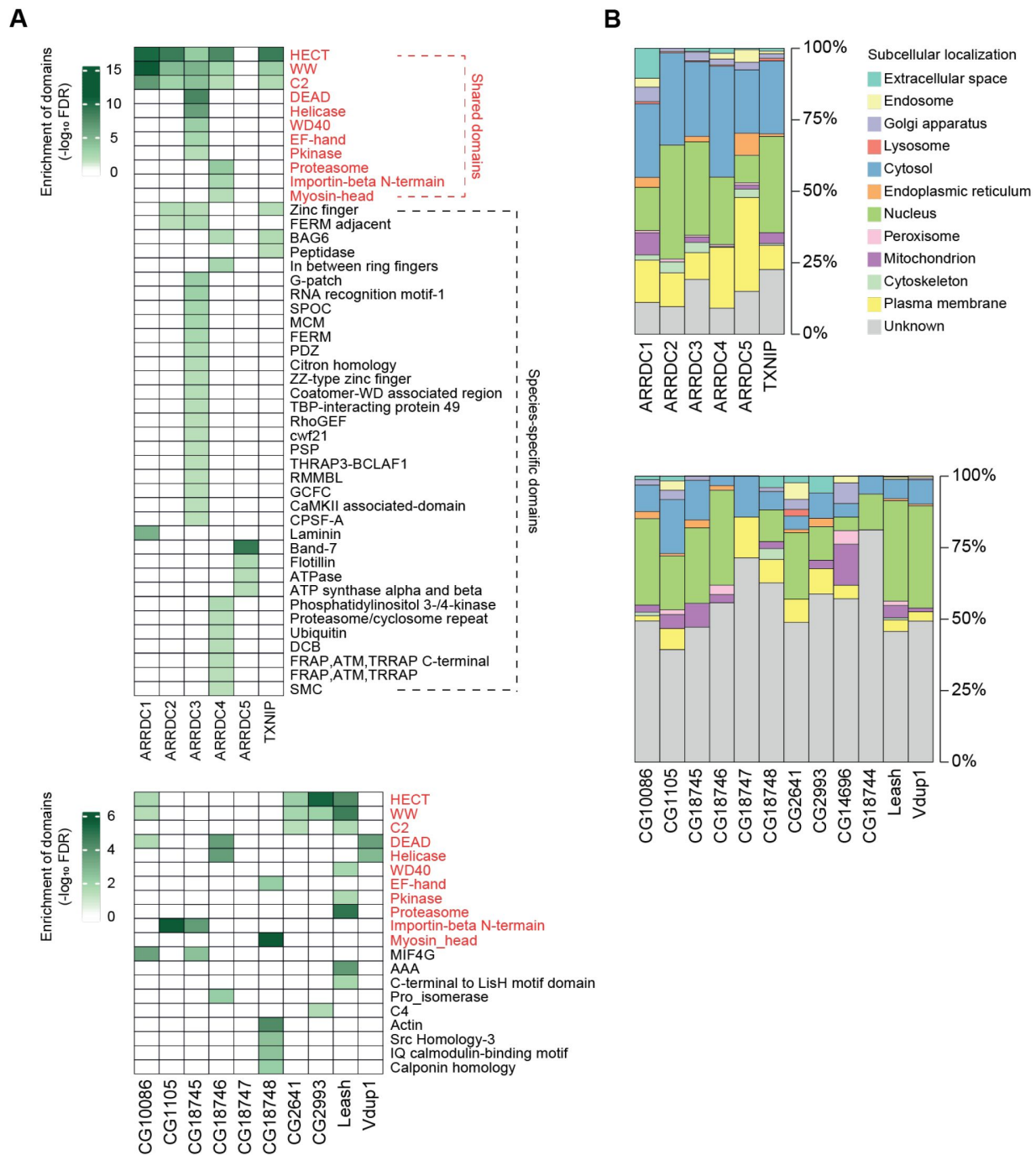


Figure S3.

Protein domains and subcellular localization of α -arrestin interactomes

(A) Protein domains enriched in each α -arrestin interactome for human (top) and *Drosophila* (bottom) are shown. The significance of the enrichment test ($-\log_{10}$ FDR) is indicated in shades of green, as depicted in the legend. SPOC, spen paralogue and orthologue C-terminal; MCM, minichromosome maintenance protein complex; FDRM, F for 4.1 protein, E for ezrin, R for radixin and M for moesin; TBP, TATA binding protein; GEF, guanine nucleotide exchange factor; THRAPP3, thyroid hormone receptor-associated protein 3; BCLAF1, Bcl-2-associated transcription factor1; RMMBL, RNA metabolising metallo beta lactamase; CaMKII, C-terminus of the Calcium/calmodulin dependent protein kinases II; CPSF, cleavage and polyadenylation specificity factor; DCB, dimerization and cyclophilin-binding domain; FRAP, FKBP12-rapamycin complex-associated protein; ATM, ataxia telangiectasia mutant; THRAPP, transformation/transcription domain associated proteins; MIF4G, middle domain of eukaryotic initiation factor 4G; AAA, ATPase family associated with various cellular activities; C4, C terminal tandem repeated domain in type 4 procollagen; SMC, structural maintenance of chromosomes.

(B) Subcellular localizations of prey proteins of each α -arrestin for human (left) and *Drosophila* (right).

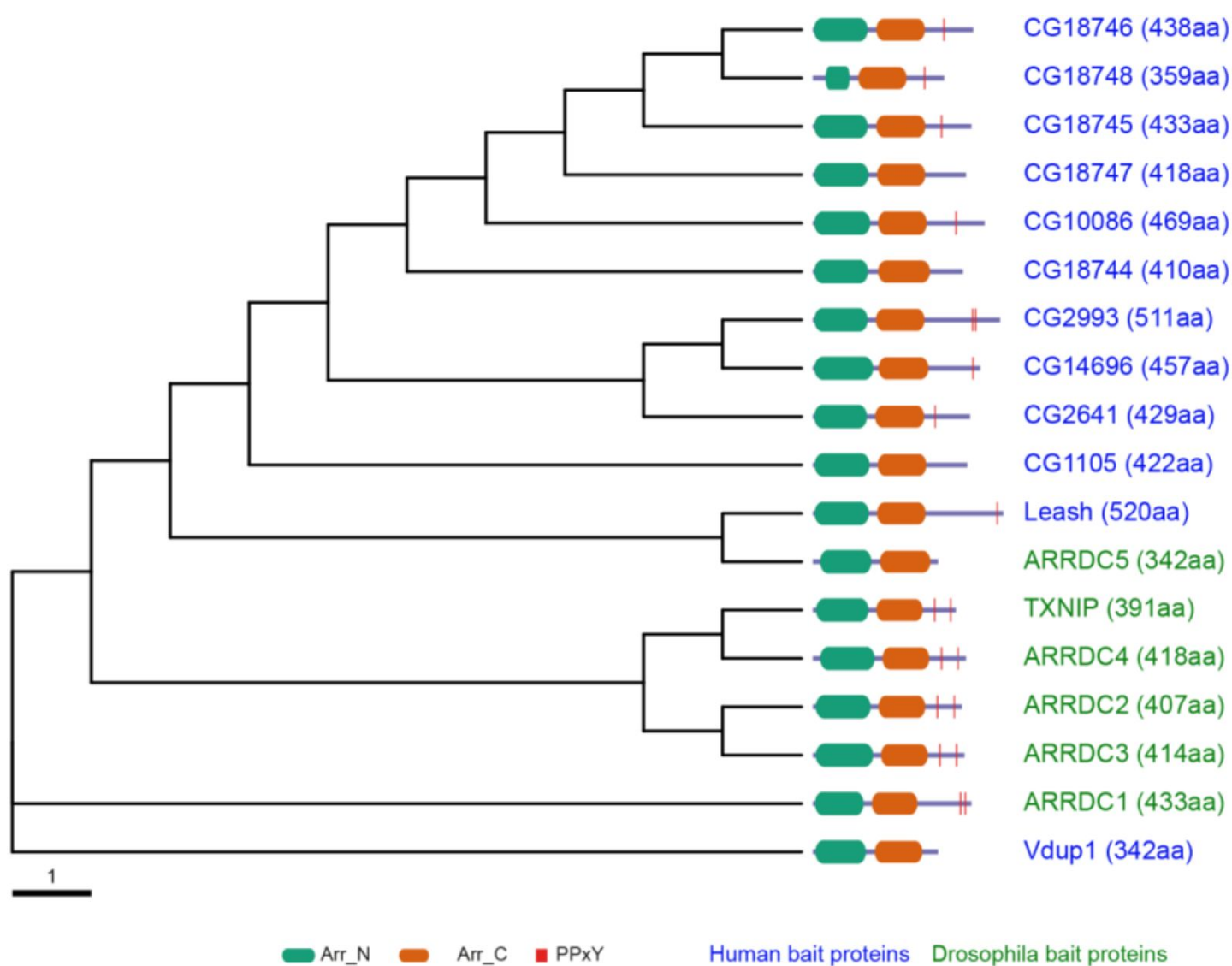


Figure S4.

Phylogenetic tree showing relationships between α -arrestins from human and *Drosophila*

Phylogenetic tree of α -arrestins from both species based on protein sequences were drawn as in [Figure 1A](#).

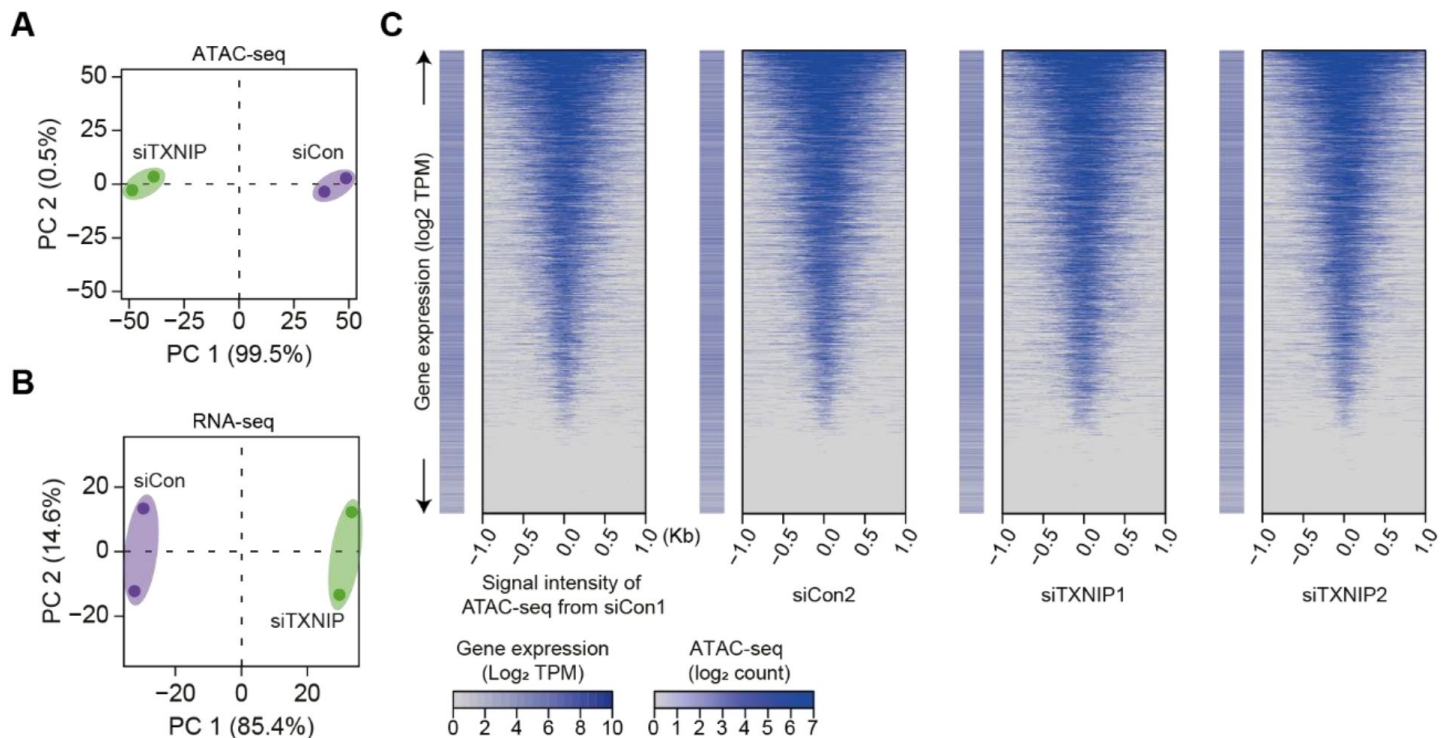


Figure S5.

ATAC- and RNA-seq results are highly reproducible and ATAC-seq results exhibit a pattern typical of strong enrichment around TSSs of expressed genes

(A and B) PCA plots of ATAC- **(A)** and RNA-seq **(B)** results based on batch-corrected log₂ counts and CPM, respectively. Numbers in parentheses are percentages of explained variance for the corresponding PCs. **(C)** Heatmaps of ATAC-seq read counts (read counts have been transformed into a log₂ function and corrected for batch effects) in regions surrounding TSSs along with log₂ (RNA level in siTXNIP-treated cells/RNA level in siCon-treated cells) for genes having the corresponding TSS are plotted for each sample.

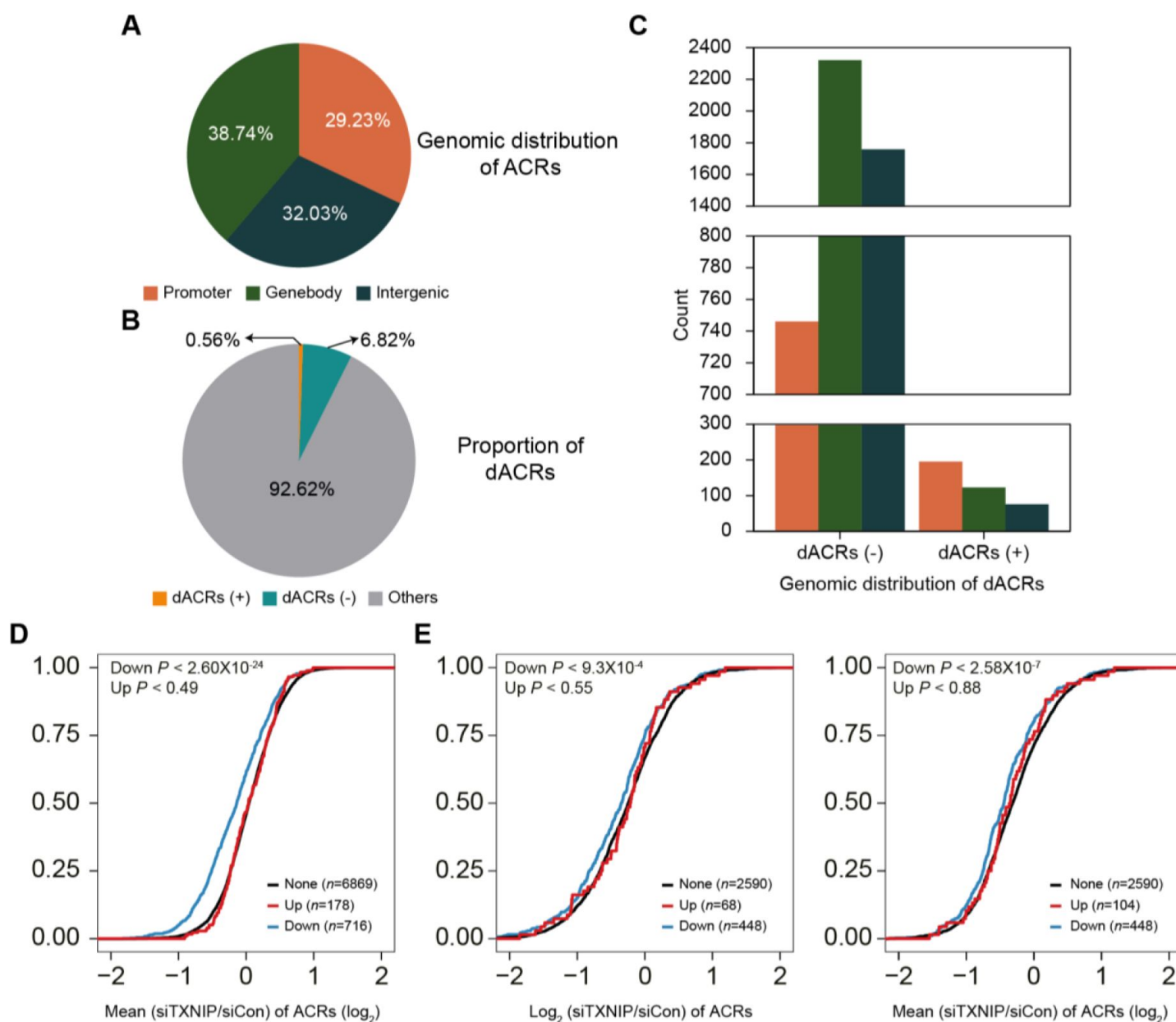


Figure S6.

Genomic locations of ACRs and association between chromatin landscapes and transcriptional activity

(A) Genomic locations of 70,746 consensus ACRs identified from ATAC-seq analysis. (B) Composition of dACRs(-), dACRs(+), and other ACRs ("others", not significantly changed) under the TXNIP knockdown condition compared to the control. (C) Genomic locations of 4,825 dACRs(-) and 394 dACRs(+) are depicted. Colors in the bar plot have the same symbolism as in (A). (D) Cumulative distribution function (CDF) of mean changes in accessibility of all ACRs located in gene promoters. The genes were categorized into three groups ("None", "Down", and "Up") as explained in Figure 4F. P values on the left upper corner were calculated with the one sided Kolmogorov-Smirnov (KS) test comparing "Down" or "Up" groups to the "None" group. (E) CDF of changes in accessibility of ACRs located in gene bodies. Changes in accessibility of ACRs whose intensity is highest among all ACRs located in gene bodies are depicted on the left and mean changes in accessibility of all ACRs located in gene bodies are depicted on the right. P values on the upper left corners are calculated in the same manner as in (D).

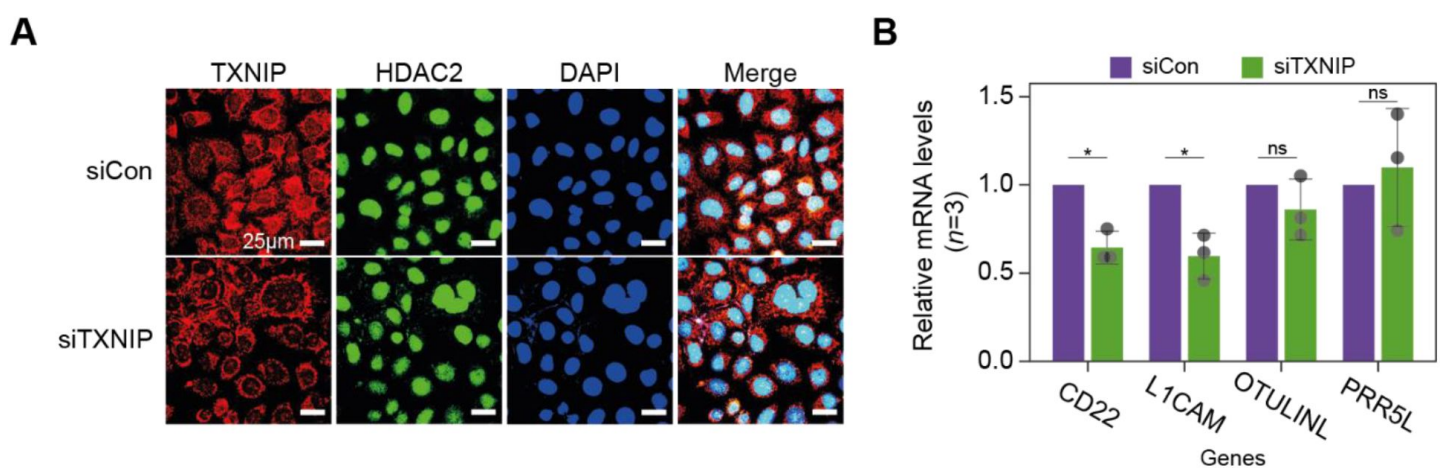


Figure S7.

TXNIP depletion does not affect the protein level or subcellular localization of HDAC2 but represses transcription of target genes

(**A**) Representative immunofluorescence images of TXNIP and HDAC2 after HeLa cells were transfected with either siCon or siTXNIP for 48 hr (magnification $\times 600$); TXNIP (red), HDAC2 (green), and DAPI (blue). (**B**) RT-qPCR results of four target genes whose RNA expression and chromatin accessibility in their promoters, quantified using high-throughput sequencing data, were observed to be strongly repressed in HeLa cell. Data are presented as the mean $\pm$ sd, n=3). Gray dots depict actual values of each experiment. *P < 0.05, ns: not significant (two sided paired Student T test).

Table S10. Result of analysis of differential accessibility of ACRs **(A)** and differential gene expression **(B)**.

Table S11. (A) Profiles of ATAC-seq peaks located in promoters of genes. Changes in peak intensities and gene expression levels are also summarized. **(B)** List of genes that exhibited decreased chromatin accessibility at their promoter and decreased RNA expression upon TXNIP knockdown.

	ATAC-seq			RNA-seq	
Sample	Properly paired reads (%)	Filtered/dedup reads*	Called peaks	Filtered reads (%)	Alignable reads
siNegative1	117,217,586 (97.8%)	17,929,628	74,373	41,756,384 (99.8%)	37,548,784
siNegative2	203,055,772 (97.7%)	31,497,080	141,799	41,900,786 (99.4%)	36,139,515
siTXNIP1	123,045,656 (97.8%)	20,050,776	69,431	41,729,984 (99.8%)	37,185,131
siTXNIP2	179,673,798 (98%)	25,159,908	125,301	39,503,312 (99.5%)	33,418,535

Table S9.

Summary of ATAC- and RNA-seq read counts before and after processing. For ATAC-seq, the number of properly paired reads, filtered/deduplicated reads, and identified narrow peaks are summarized. For RNA-seq, the number of filtered and alignable reads are summarized. *Filtered/dedup reads, filtered/deduplicated reads

Primer name	Forward/reverse	Sequence	Application
alpha-tubulin	Forward	CTGGACCGCATCTCTGTGTACT	RT-qPCR
	Reverse	GCCAAAAGGACCTGAGCGAACA	
TXNIP	Forward	GCTCCTCCCTGCTATATGGAT	
	Reverse	AGTATAAGTCGGTGGTGGCAT	
CD22	Forward	GCGCAGCTTGTAAATAGTTGGTGC	
	Reverse	CACATTGGAGGCTGACCGAGTT	
L1CAM	Forward	TCGCCCTATGTCCACTACACCT	
	Reverse	ATCCACAGGGTTCTTCTCTGGG	
CD22	Forward	GCGCAGCTTGTAAATAGTTGGTGC	
	Reverse	CACATTGGAGGCTGACCGAGTT-	
OTULINL	Forward	GTGTGGAGGCAGAGGTTGAT	
	Reverse	ATGCCGCCAAAATAGCTCCT	
PRR5L	Forward	GCGGCTGTTGAAGAGTGAAC	
	Reverse	AGCCAGAACCTCAATGCGAT	
SDC3	Forward	CTCCTGGACAATGCCATCGACT	
	Reverse	TGAGCAGTGTGACCAAGAAGGC	
GAPDH	Forward	ATCACCATCTTCCAGGAGCGA	ChIP-qPCR
	Reverse	CCTTCTCCATGGTGGTGAAGAC	
CD22 #1	Forward	CGCTGGAGAAGTGAGTTCGG	
	Reverse	TCCCTGCCTCCACTGATAGC	
CD22 #2	Forward	GACGCTGAGATGAGGGTTGG	
	Reverse	TGACTCAGGAGGTTGGCAGA	
CD22 #3	Forward	TCCCCACTCTTCTCGCTCTC	
	Reverse	ATTTGCGAGGTTGAGGTTGTC	
L1CAM #1	Forward	CAGCTCAGTGCCTCATGGAA	
	Reverse	GAGACTGCTTCCAGAGTGGG	
CD22 #2	Forward	GGAATGCTTCACTGGGCAAC	
	Reverse	GGGGTAAGAATTCCGGAGCC	
CD22 #3	Forward	CGTGTCTGAGAAAGGAAGCCA	
	Reverse	CGGCTTATCCCGATCTACCC	

Table S12.

List of primer sequences used in this study.

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Reviewer #1 (Public Review):

The study provides a complete comparative interactome analysis of α -arrestin in both humans and drosophila. The authors have presented interactomes of six humans and twelve *Drosophila* α -arrestins using affinity purification/mass spectrometry (AP/MS). The constructed interactomes helped to find α -arrestins binding partners through common protein motifs. The authors have used bioinformatic tools and experimental data in human cells to identify the roles of TXNIP and ARRDC5: TXNIP-HADC2 interaction and ARRDC5-V-type ATPase interaction. The study reveals the PPI network for α -arrestins and examines the functions of α -arrestins in both humans and *Drosophila*.

Comments

I will like to congratulate the authors and the corresponding authors of this manuscript for bringing together such an elaborate study on α -arrestin and conducting a comparative study in *Drosophila* and humans.

Introduction:

The introduction provides a rationale behind why the comparison between humans and *Drosophila* is carried out.

- Even though this is a research manuscript, including existing literature on similar comparison of α -arrestin from other articles will invite a wide readership.

Results:

The results cover all the necessary points concluded from the experiments and computational analysis.

- The authors could point out the similarity of the α -arrestin in both humans and *Drosophila*.
- Citing the direct connecting genes from the network in the text will invite citations and a wider readership.

Figures:

The images are elaborate and well-made.

- The authors could use a direct connected gene-gene network that pointing interactions. This can be used by other readers working on the same topic and ensure reproducibility and citations.
- The blot/gel images can be of higher resolution.

Discussion:

The authors have utilized and discussed the conclusion they draw from their study. But could highlight more on ARRDCs and why it was selected out of the other arrestins. The authors have provided future work directions associated with their work.

Supplementary figures:

The authors have a rigorous amount of work added together for the success of this manuscript.

Reviewer #2 (Public Review):

In this manuscript, the authors present a novel interactome focused on human and fly α -arrestin family proteins and demonstrate its application in understanding the functions of these proteins. Initially, the authors employed AP/MS analysis, a popular method for mapping protein-protein interactions (PPIs) by isolating protein complexes. Through rigorous statistical and manual quality control procedures, they established two robust interactomes, consisting of 6 baits and 307 prey proteins for humans, and 12 baits and 467 prey proteins for flies. To gain insights into the gene function, the authors investigated the interactors of α -arrestin proteins through various functional analyses, such as gene set enrichment. Furthermore, by comparing the interactors between humans and flies, the authors described both conserved and species-specific functions of the α -arrestin proteins. To validate their findings, the authors performed several experimental validations for TXNIP and ARRDC5 using ATAC-seq, siRNA knockdown, and tissue staining assays. The experimental results strongly support the predicted functions of the α -arrestin proteins and underscore their importance.

Reviewer #3 (Public Review):

Lee, Kyungtae and colleagues have discovered and mapped out α -arrestin interactomes in both human and *Drosophila* through the affinity purification/mass spectrometry and the

SAINTexpress method. They found the high confident interactomes, consisting of 390 protein-protein interactions (PPIs) between six human alpha-arrestins and 307 preproteins, as well as 740 PPIs between twelve *Drosophila* alpha-arrestins and 467 prey proteins. To define and characterize these identified alpha-arrestin interactomes, the team employed a variety of widely recognized bioinformatics tools. These included protein domain enrichment analysis, PANTHER for protein class enrichment, DAVID for subcellular localization analysis, COMPLEAT for the identification of functional complexes, and DIOPT to identify evolutionary conserved interactomes. Through these analyses, they confirmed known alpha-arrestin interactors' role and associated functions such as ubiquitin ligase and protease. Furthermore, they found unexpected biological functions in the newly discovered interactomes, including RNA splicing and helicase, GTPase-activating proteins, ATP synthase. The authors carried out further study into the role of human TXNIP in transcription and epigenetic regulation, as well as the role of ARRDC5 in osteoclast differentiation. This study holds important value as the newly identified alpha-arrestin interactomes are likely aiding functional studies of this group of proteins. Despite the overall support from data for the paper's conclusions, certain elements related to data quantification, interpretation, and presentation demand more detailed explanation and clarification.

1. In Figure 1B, it is shown that human alpha-arrestins were N-GFP tagged (N-terminal) and *Drosophila* alpha-arrestins were C-GFP (C-terminal). However, the rationale of why the authors used different tags for human and fly proteins was not explained in the main text and methods.
2. In Figure 2A, there seems to be an error for labeling the GAL4p/GAL80p complex that includes NOTCH2, NOTCH1 and TSC2.
3. In Figure 5, given that knockdown of TXNIP did not affect the levels and nuclear localization of HDAC2, the authors suggest that TXNIP might modulate HDAC2 activity. However, the ChIP assay suggest a different model - TXNIP-HDAC2 interaction might inhibit the chromatin occupancy of HDAC2, reducing histone deacetylation and increasing global chromatin accessibility. The authors need to propose a model consistent with these sets of all data.
4. The authors showed that ectopic expression of ARRDC5 increased osteoclast differentiation and function. Does loss of ARRDC5 lead to defects in osteoclast function and fate determination?
5. From Figure 6D, the authors argued that ARRDC5 overexpression resulted in more V-ATPase signals; however, there is no quantification. Quantification of the confocal images will foster the conclusion. Also, western blots for V-ATPase proteins will provide an alternative way to determine the effects of ARRDC5.
6. The results from Figure 6D did not support the authors' argument that ARRDC5 might control the membrane localization of the V-ATPase, as bafilomycin is the V-ATPase inhibitor. ARRDC5 knockdown experiments will help to determine whether ARRDC5 can control the membrane localization of the V-ATPase in osteoclast.